



# Default\_Project\_Name\_1787096444739

PROJECT DOCUMENT

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Exported on: 18 August 2026, 17:35 MST

## PROJECT DETAILS

**Project name:**

Default\_Project\_Name\_1787096444739

**Estimated Annual Units:**

1

**Generated by:**

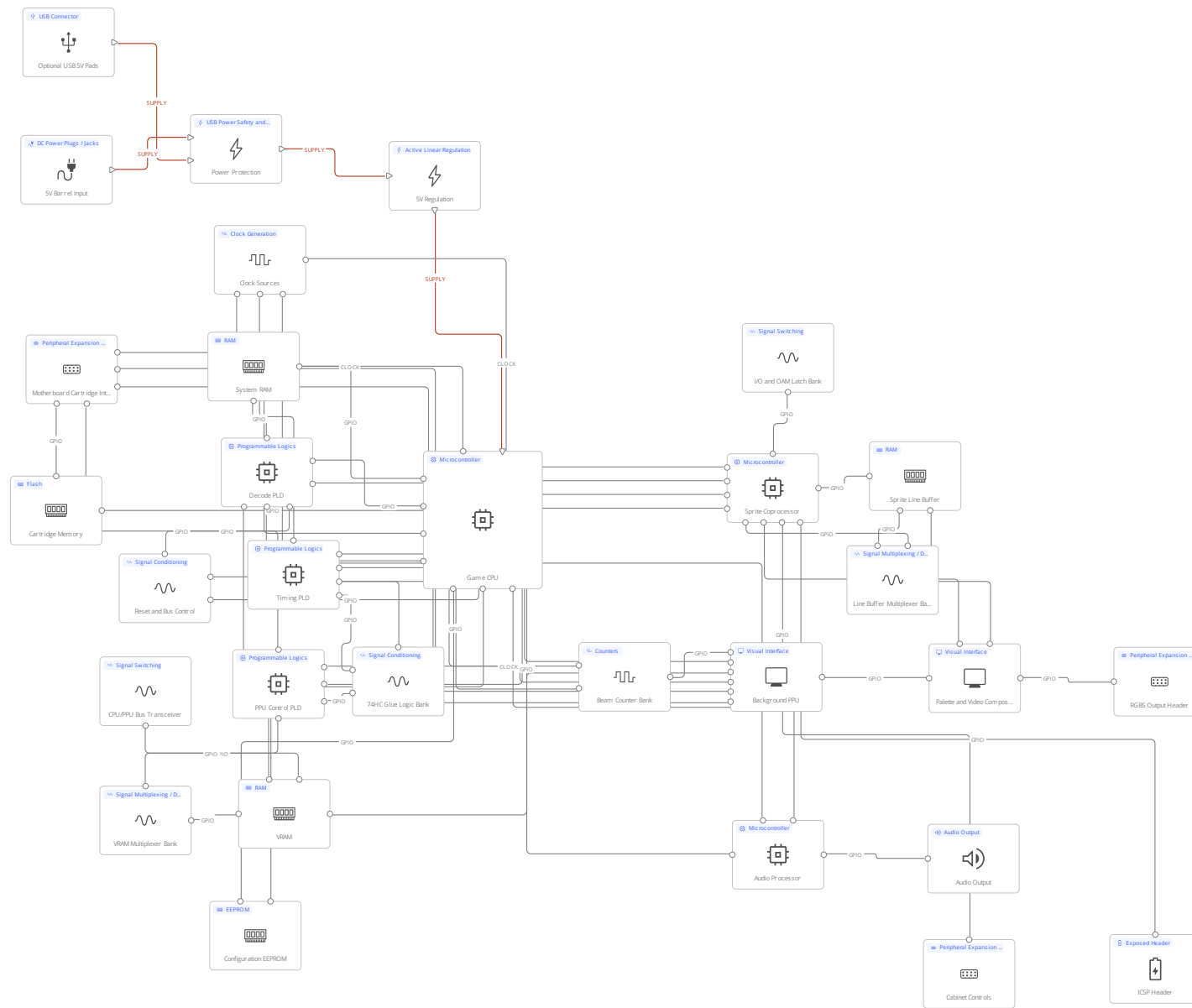
Gustavo Lara, [tavuntu@gmail.com](mailto:tavuntu@gmail.com)

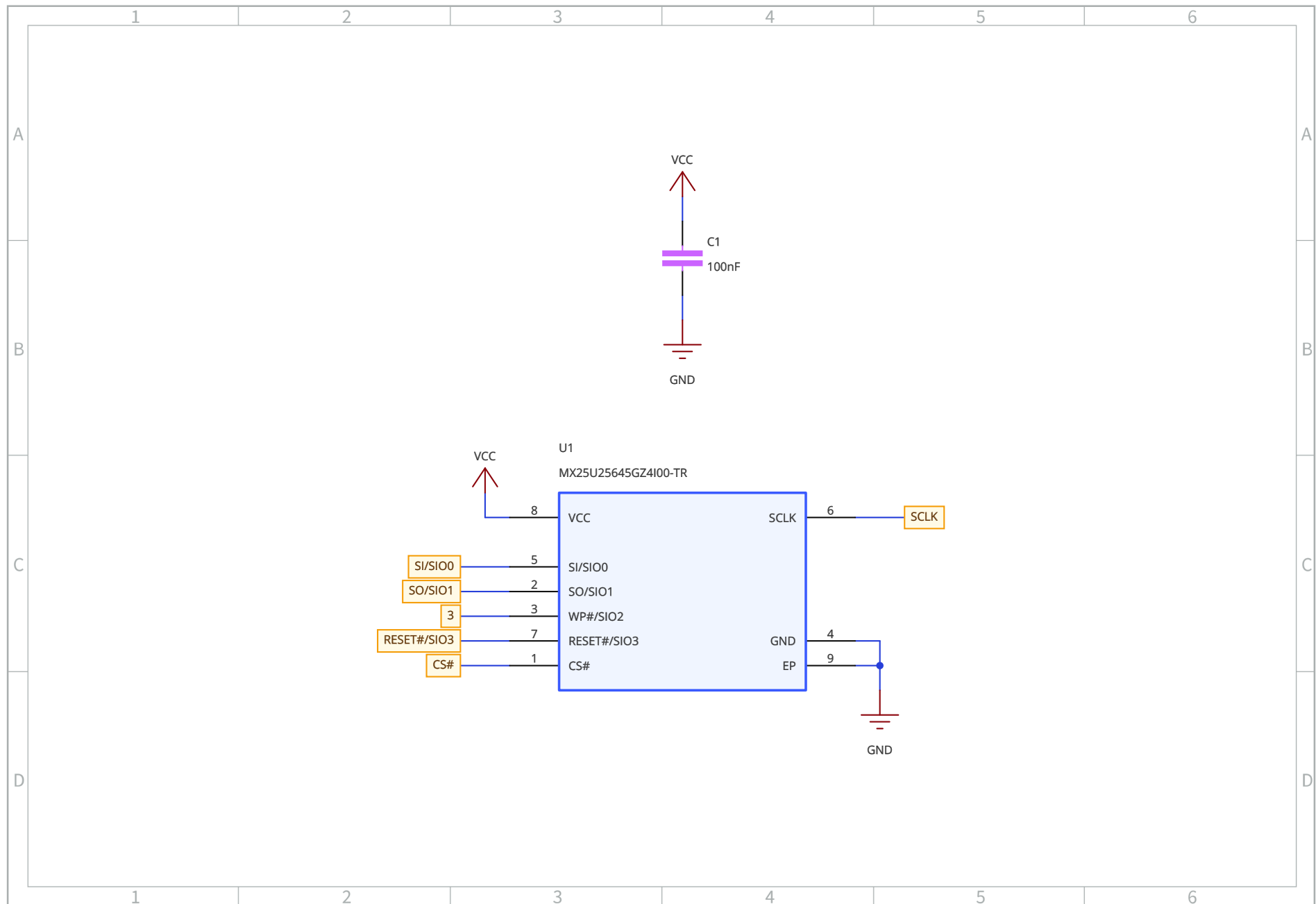
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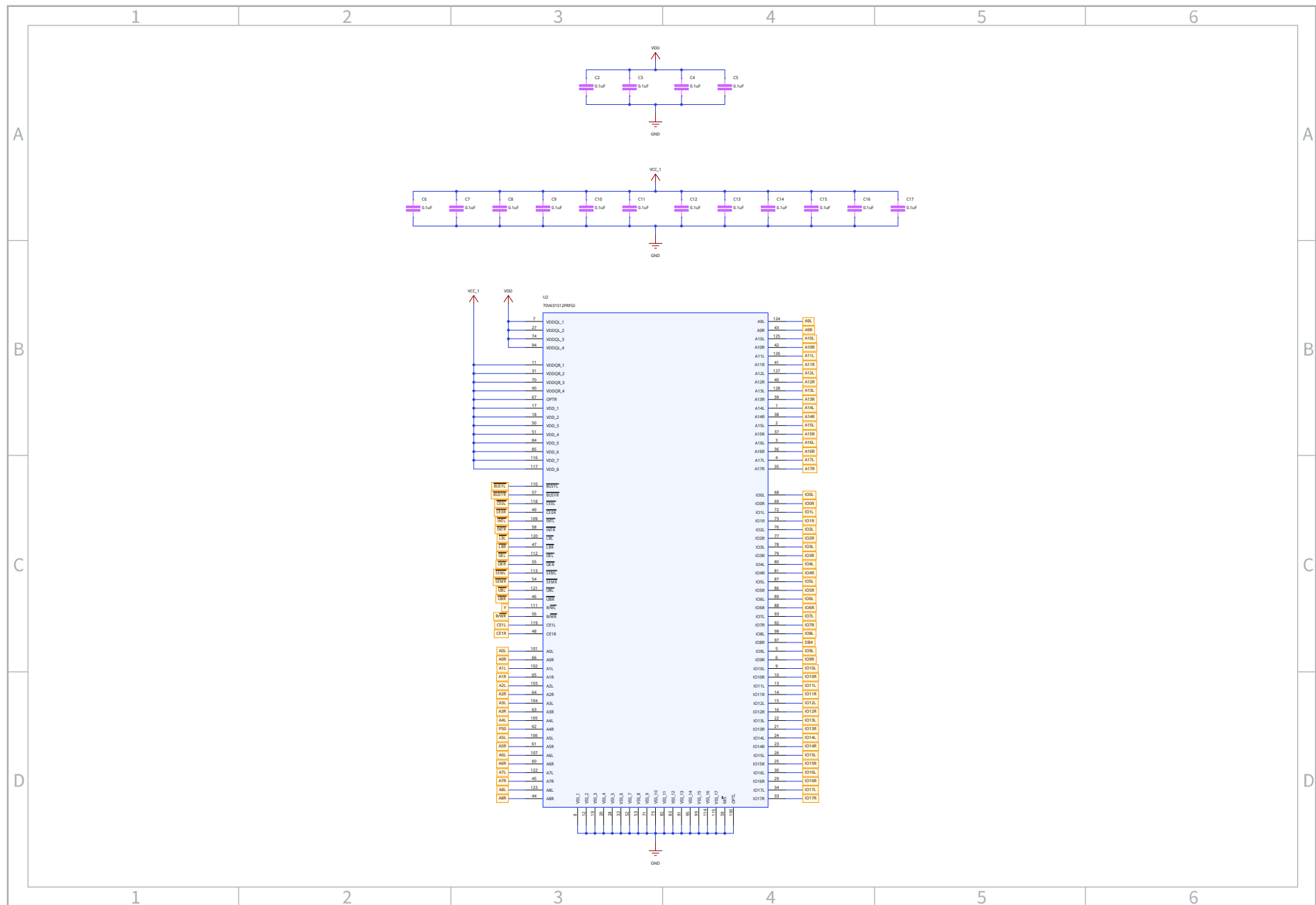
18 August 2026, 17:35 MST

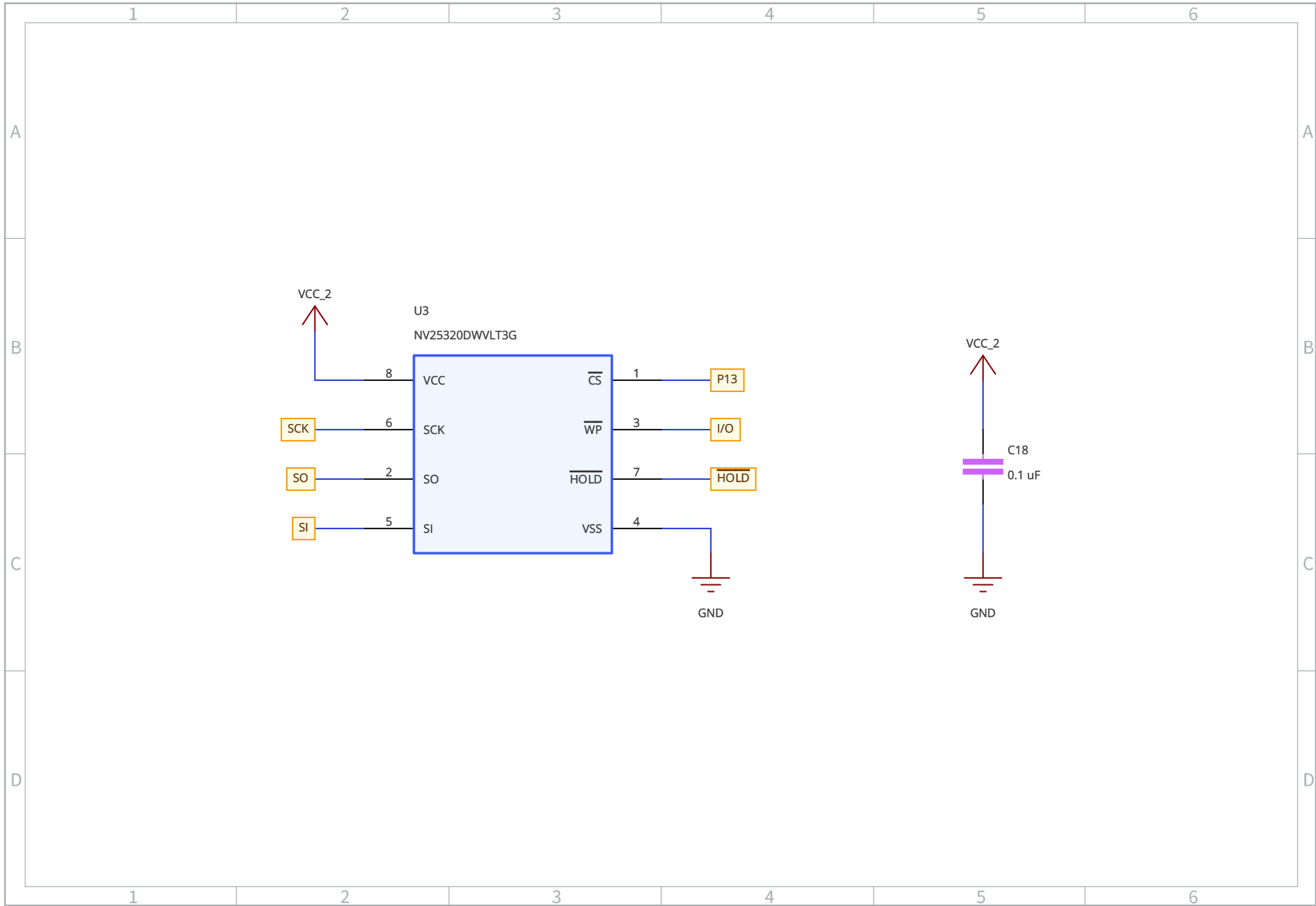
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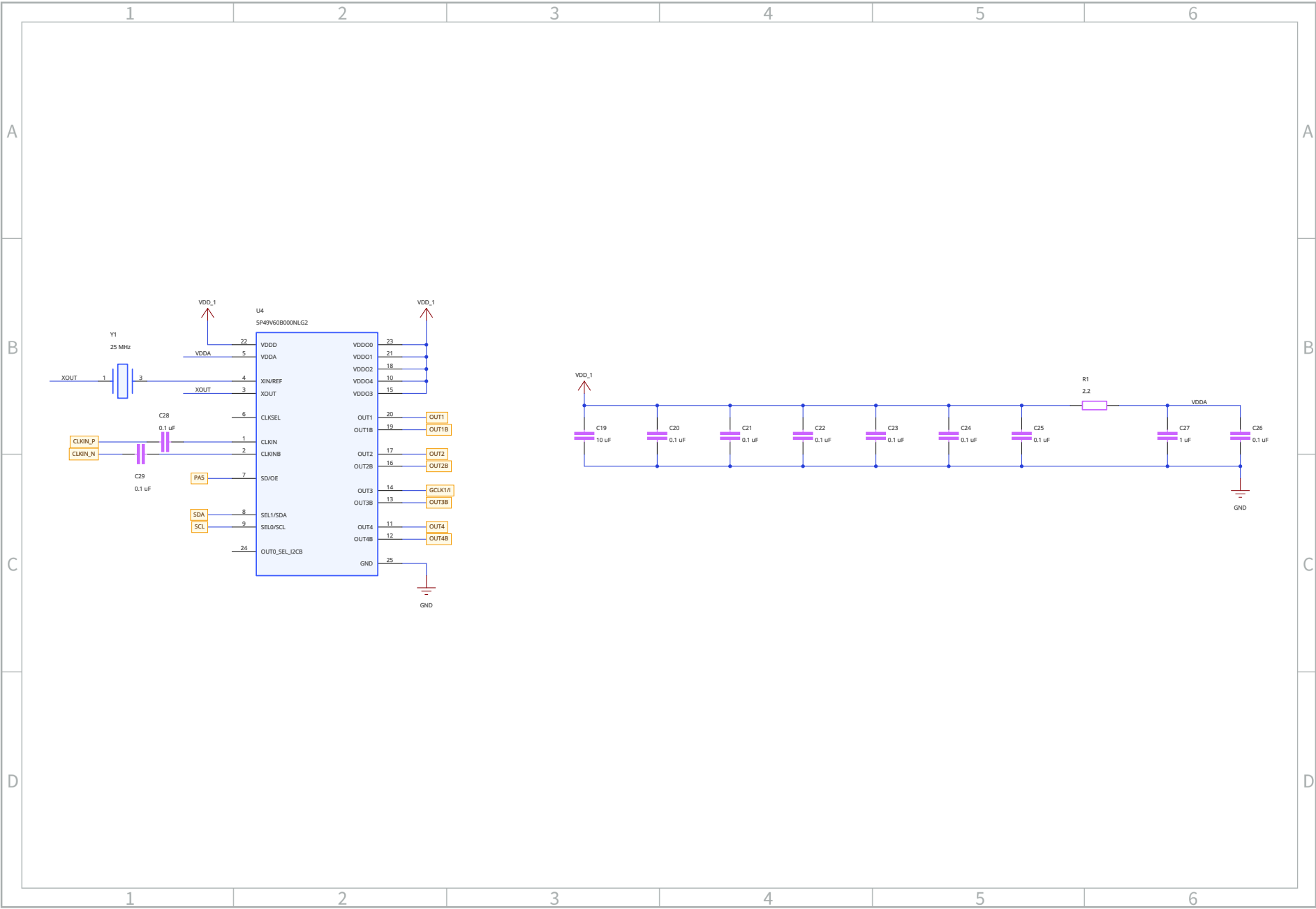
Project Title, Project Description, and Block Diagram may be AI-Generated.  
Schematics, BOM, and Components are not AI-Generated.

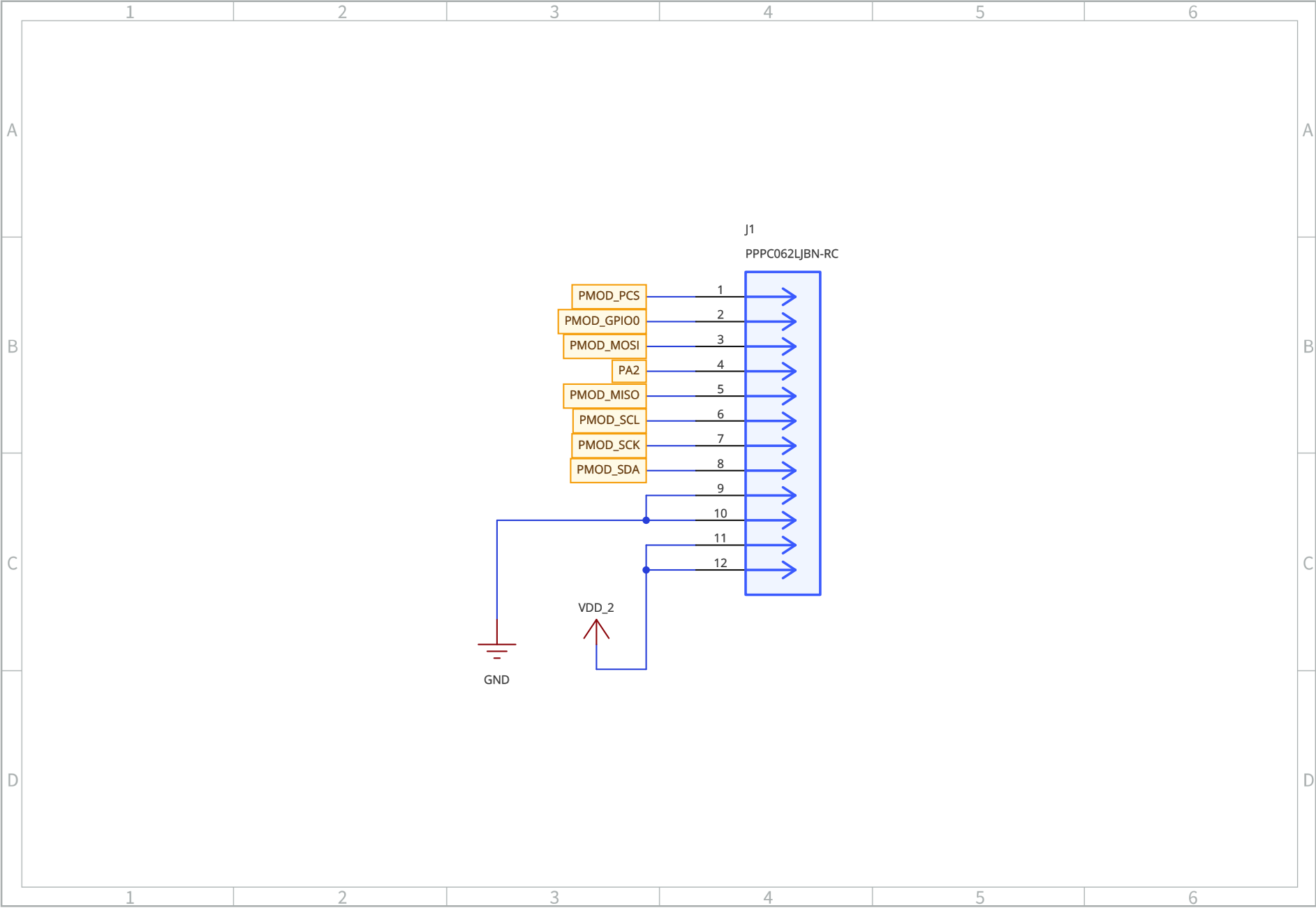






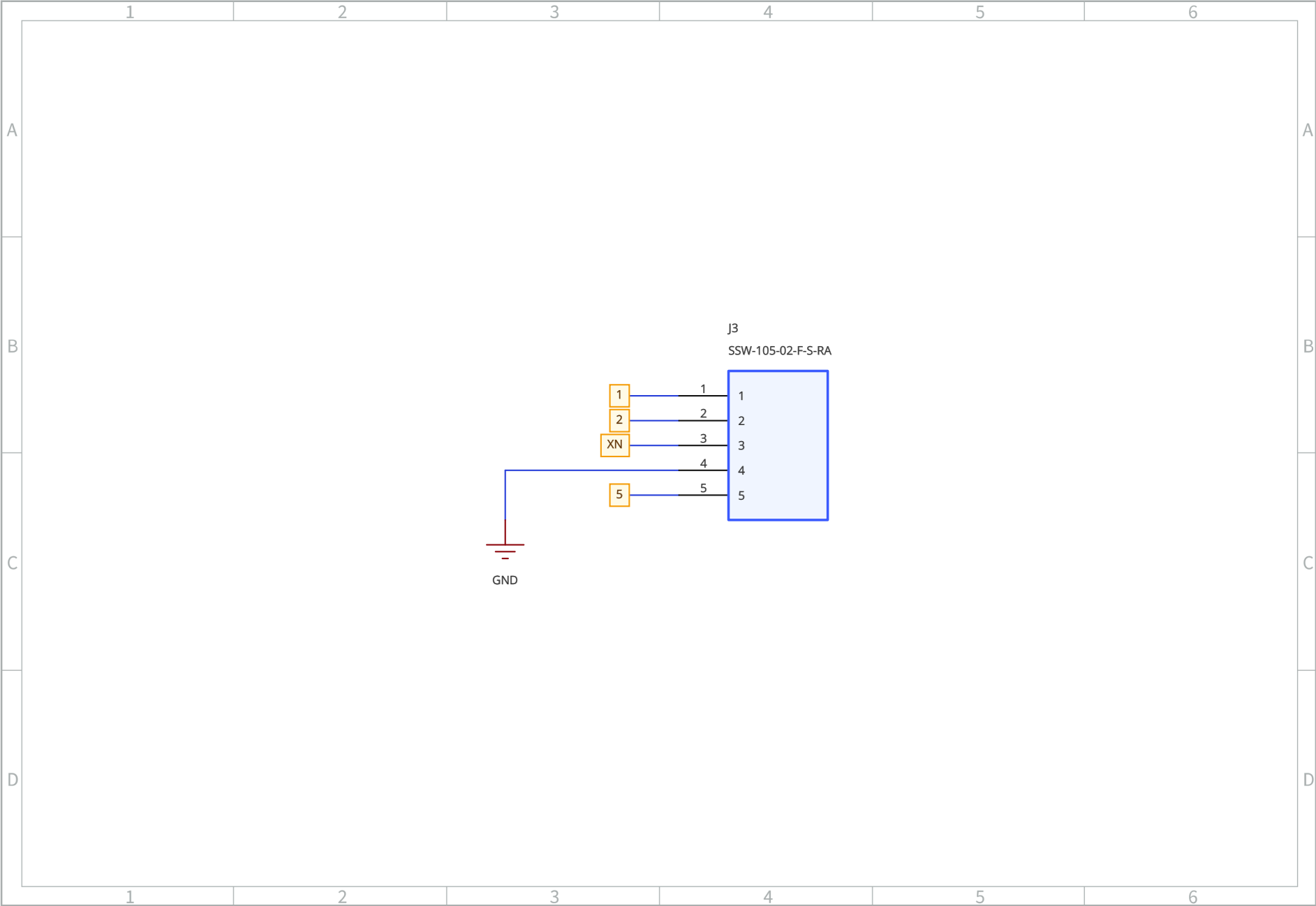


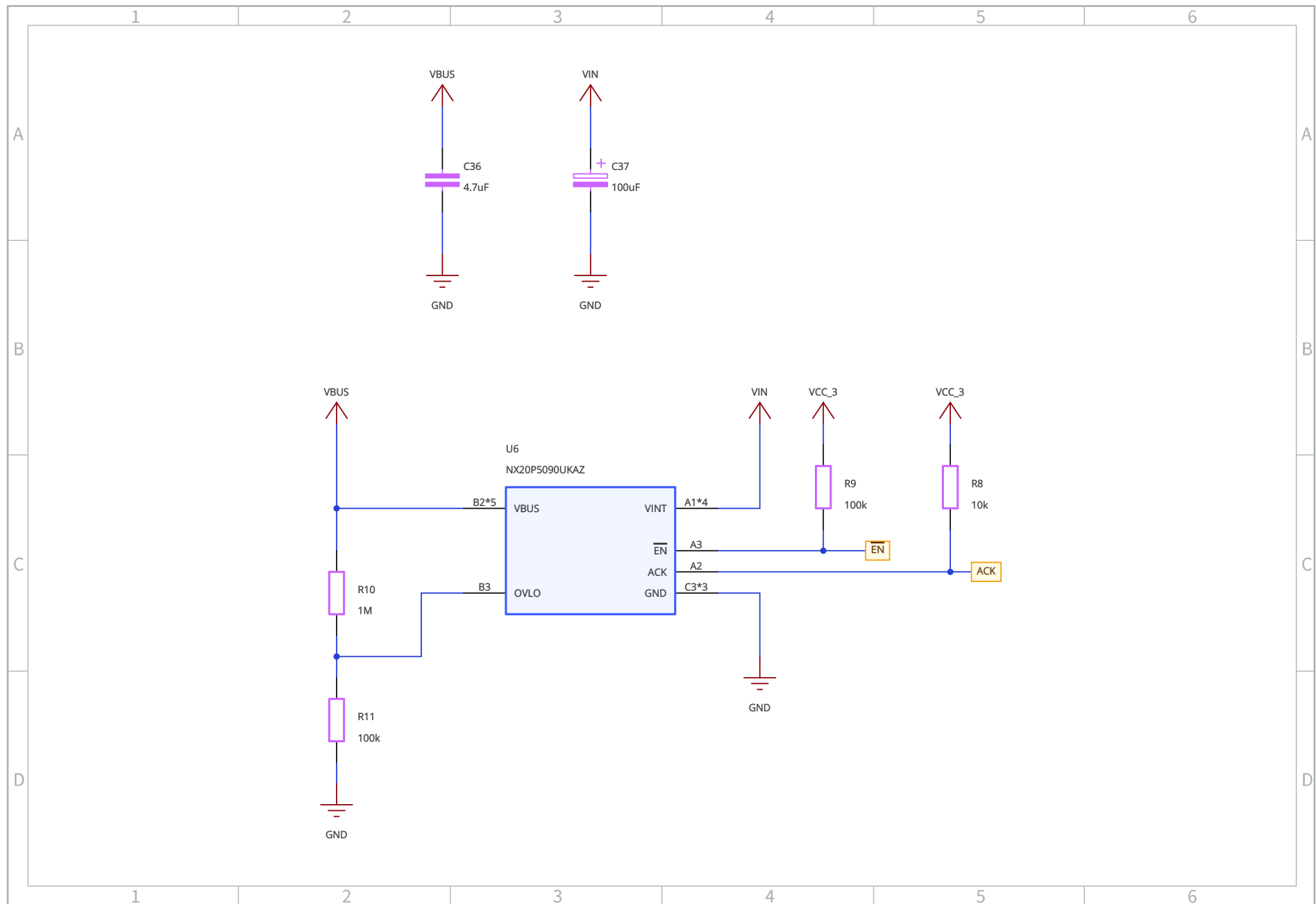


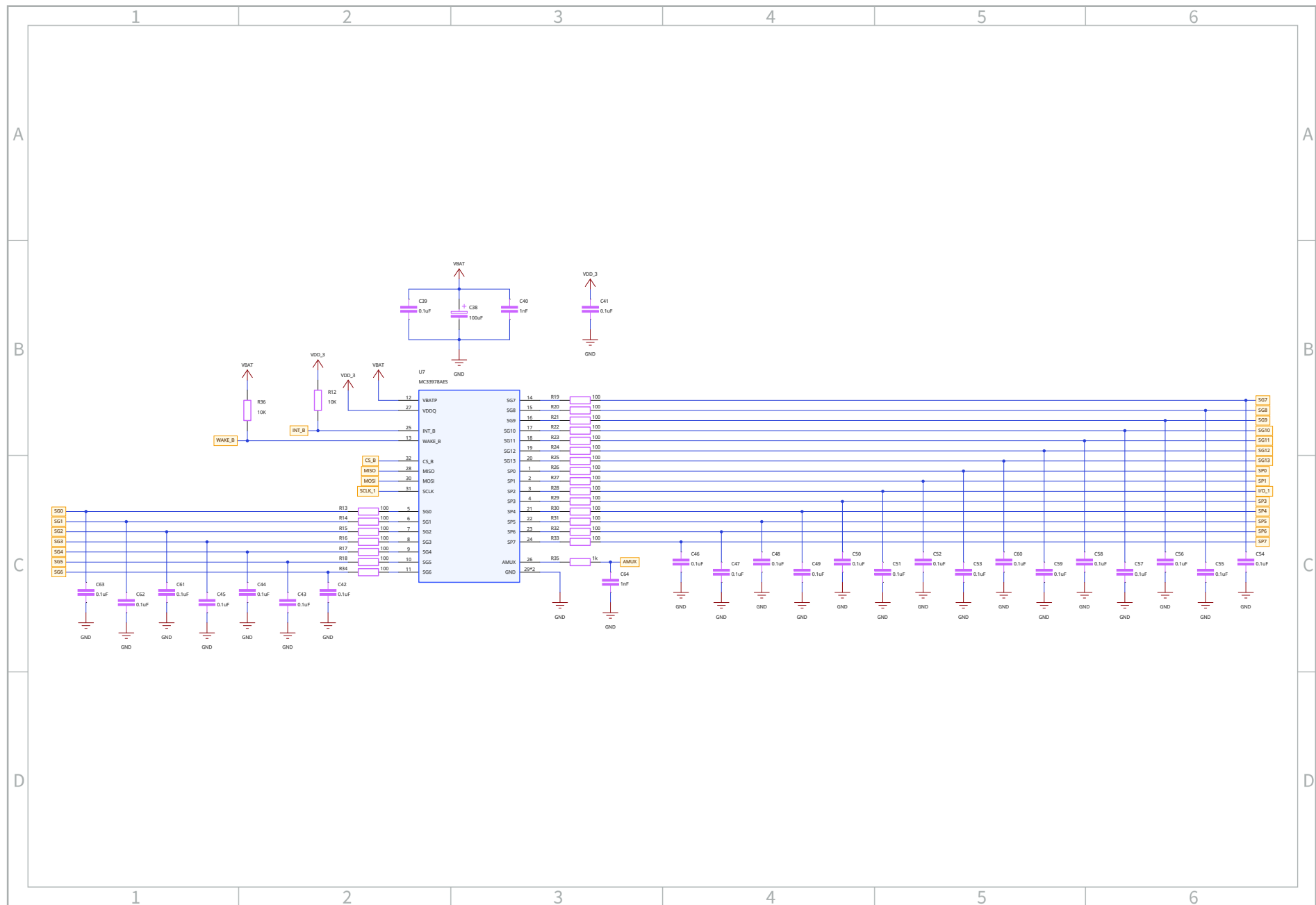


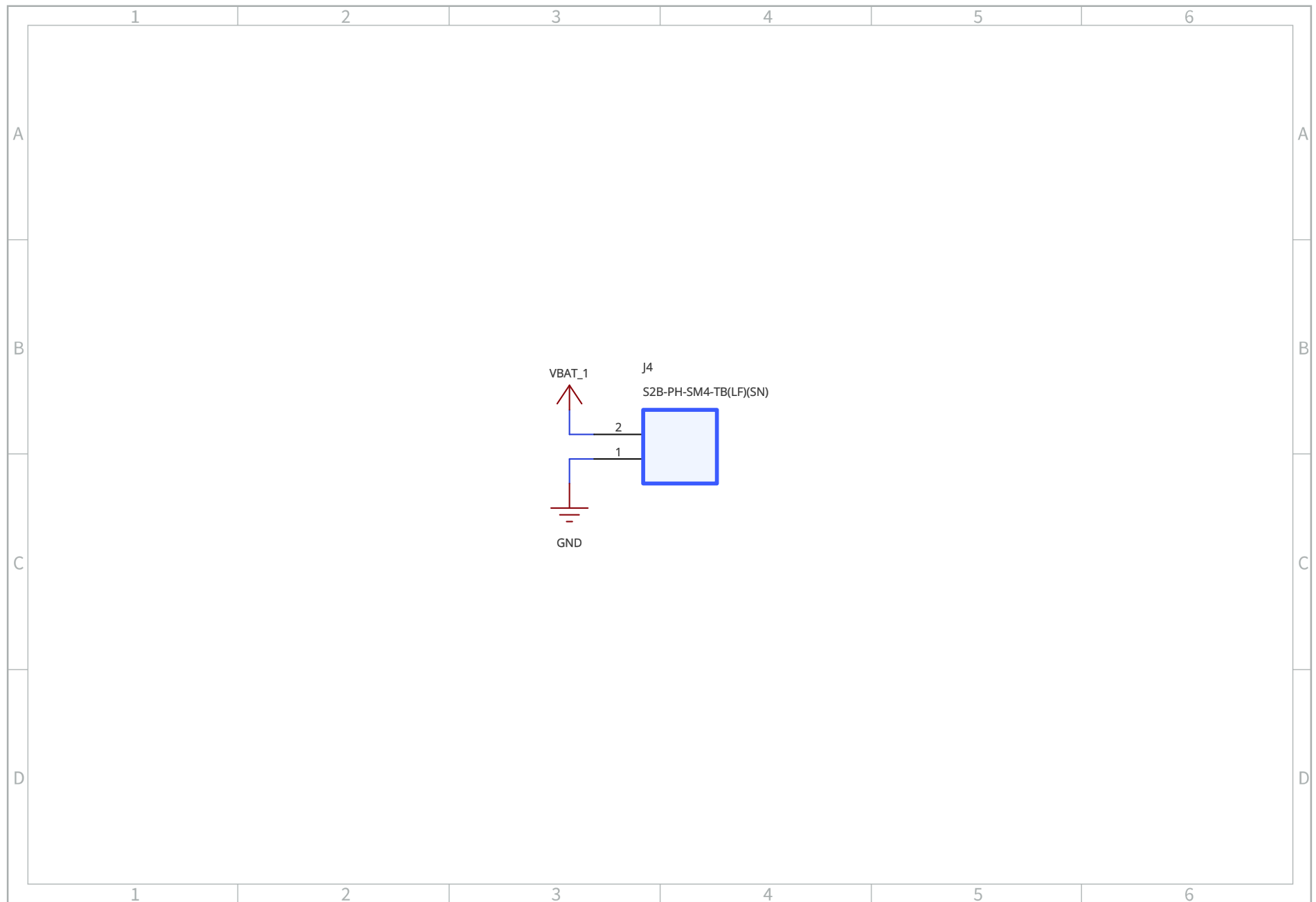


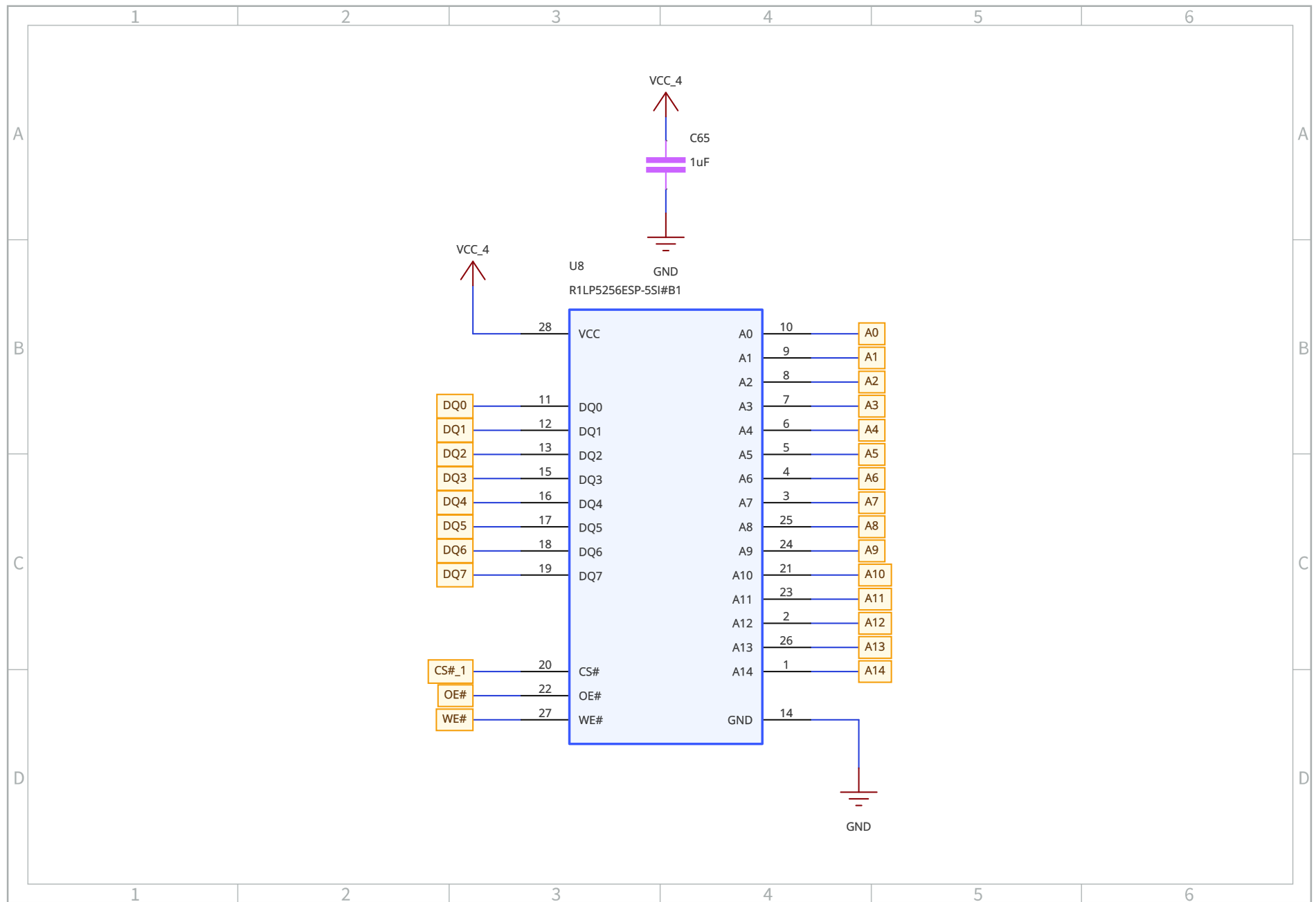


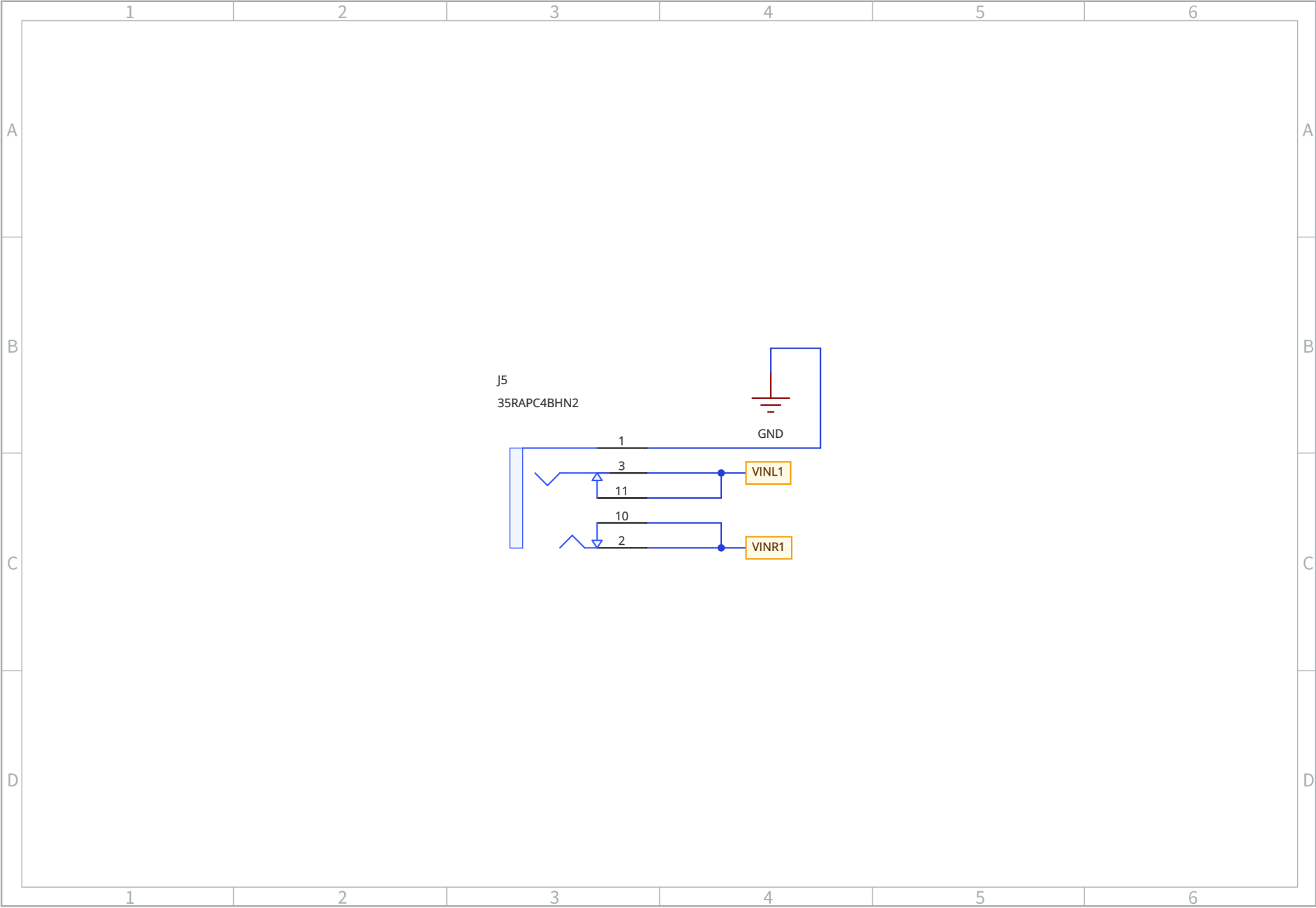


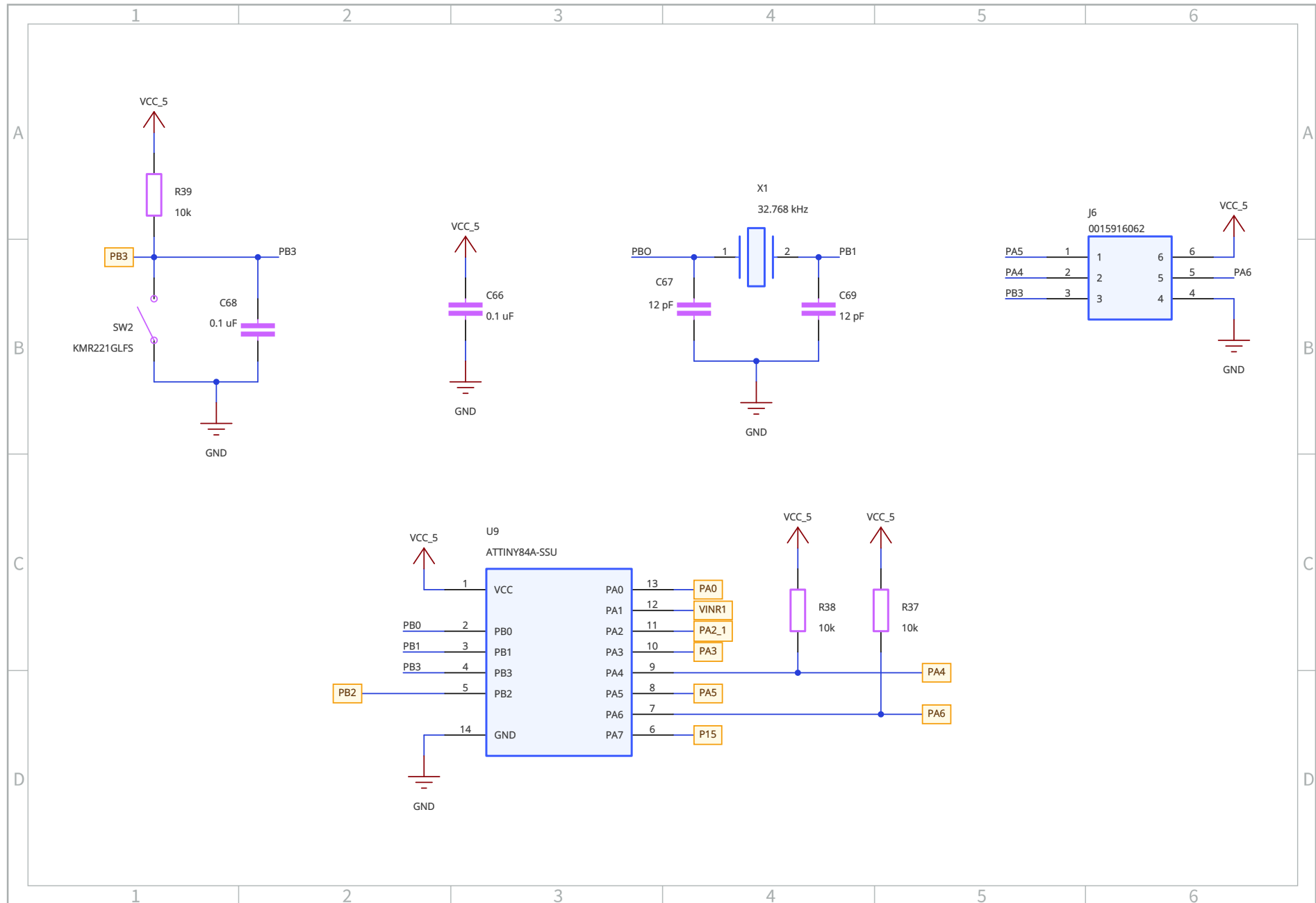




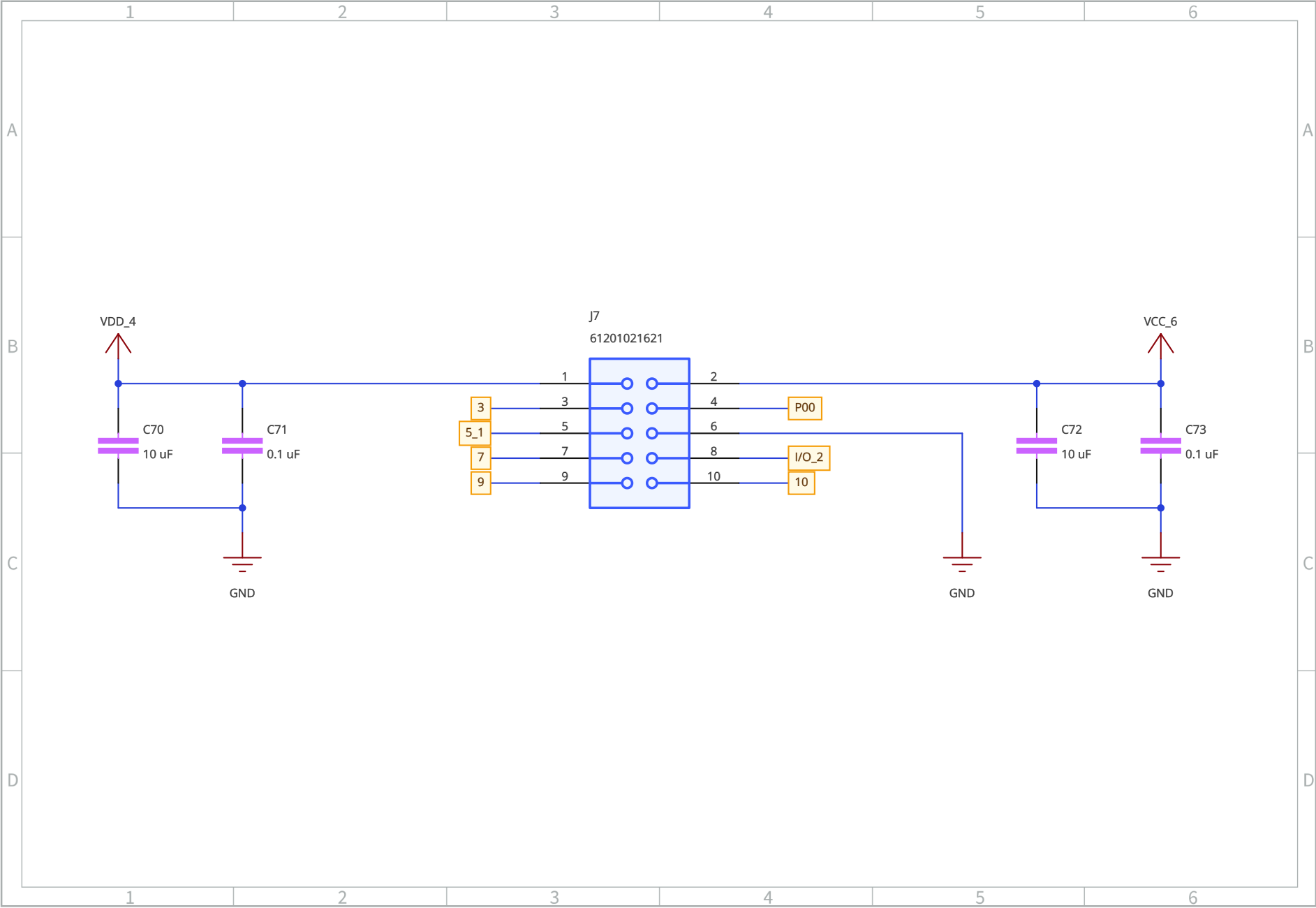


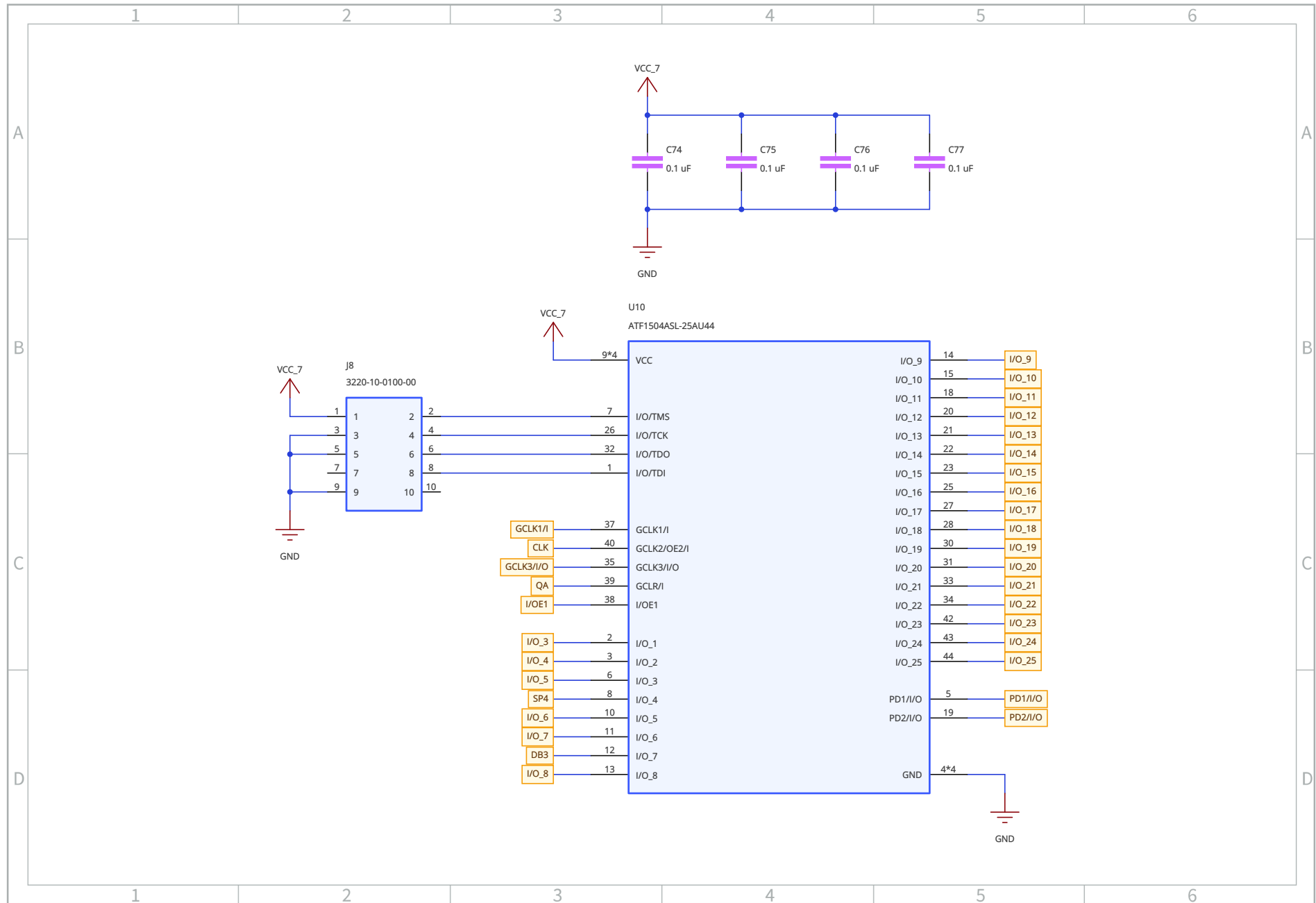


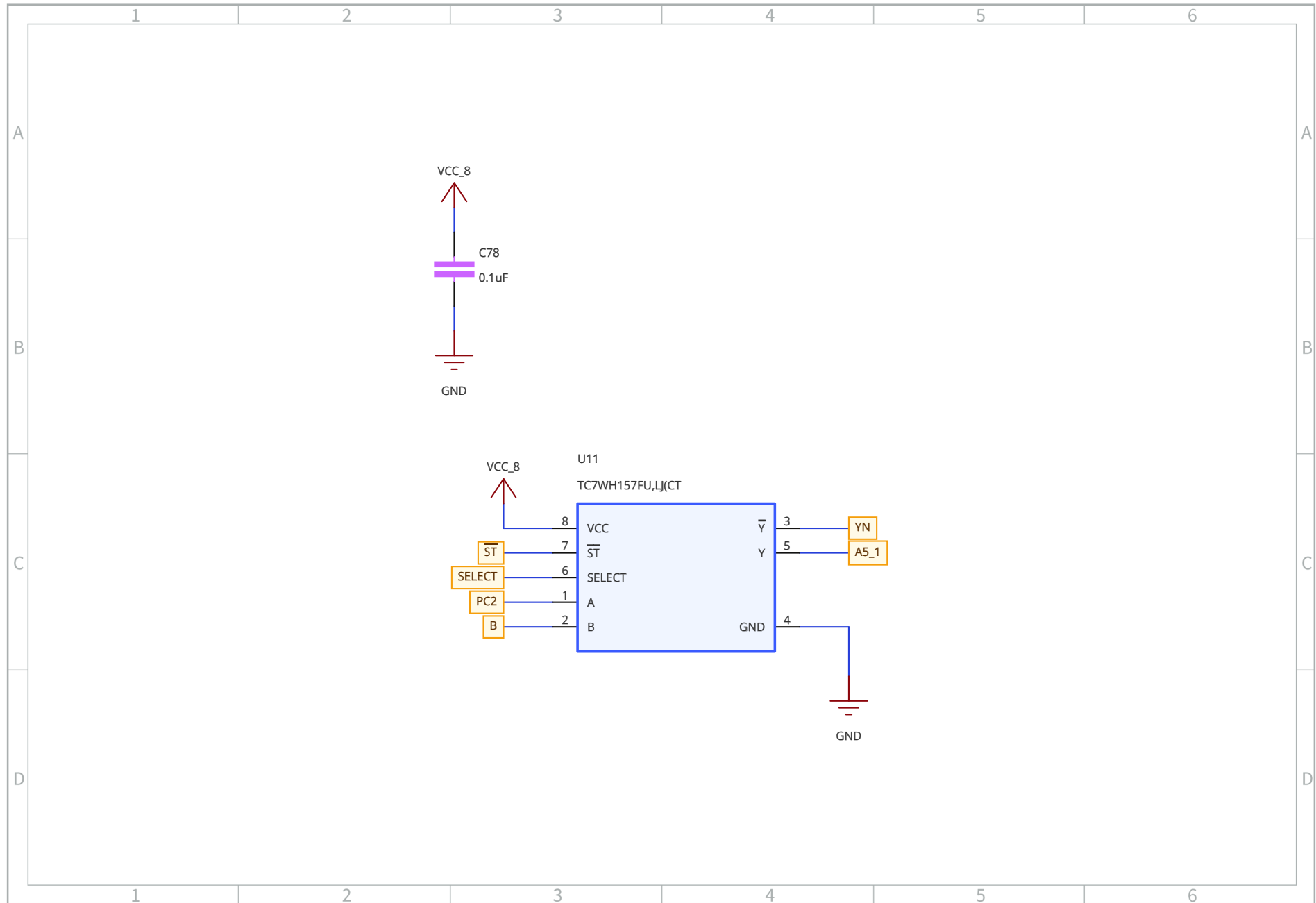


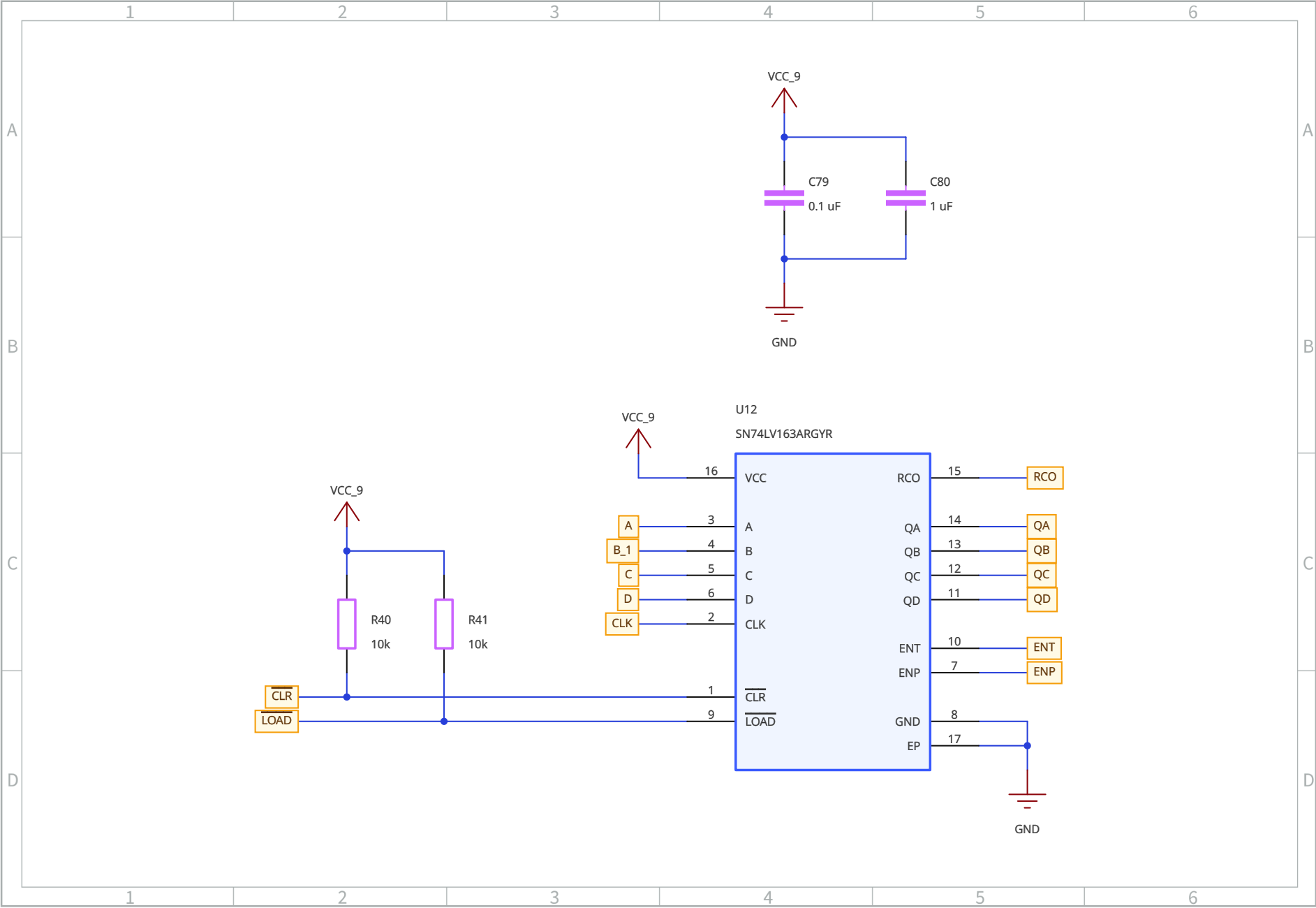


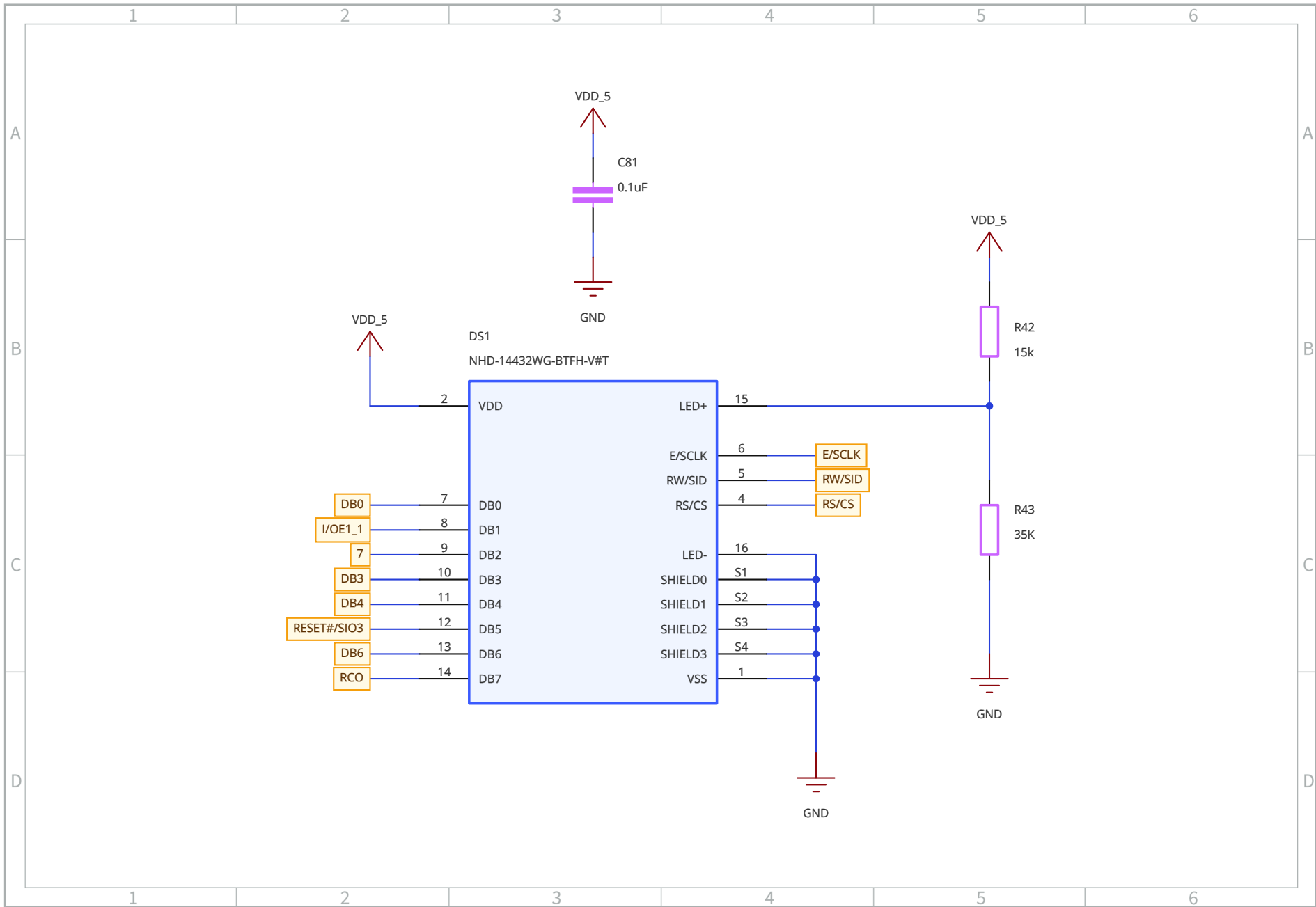


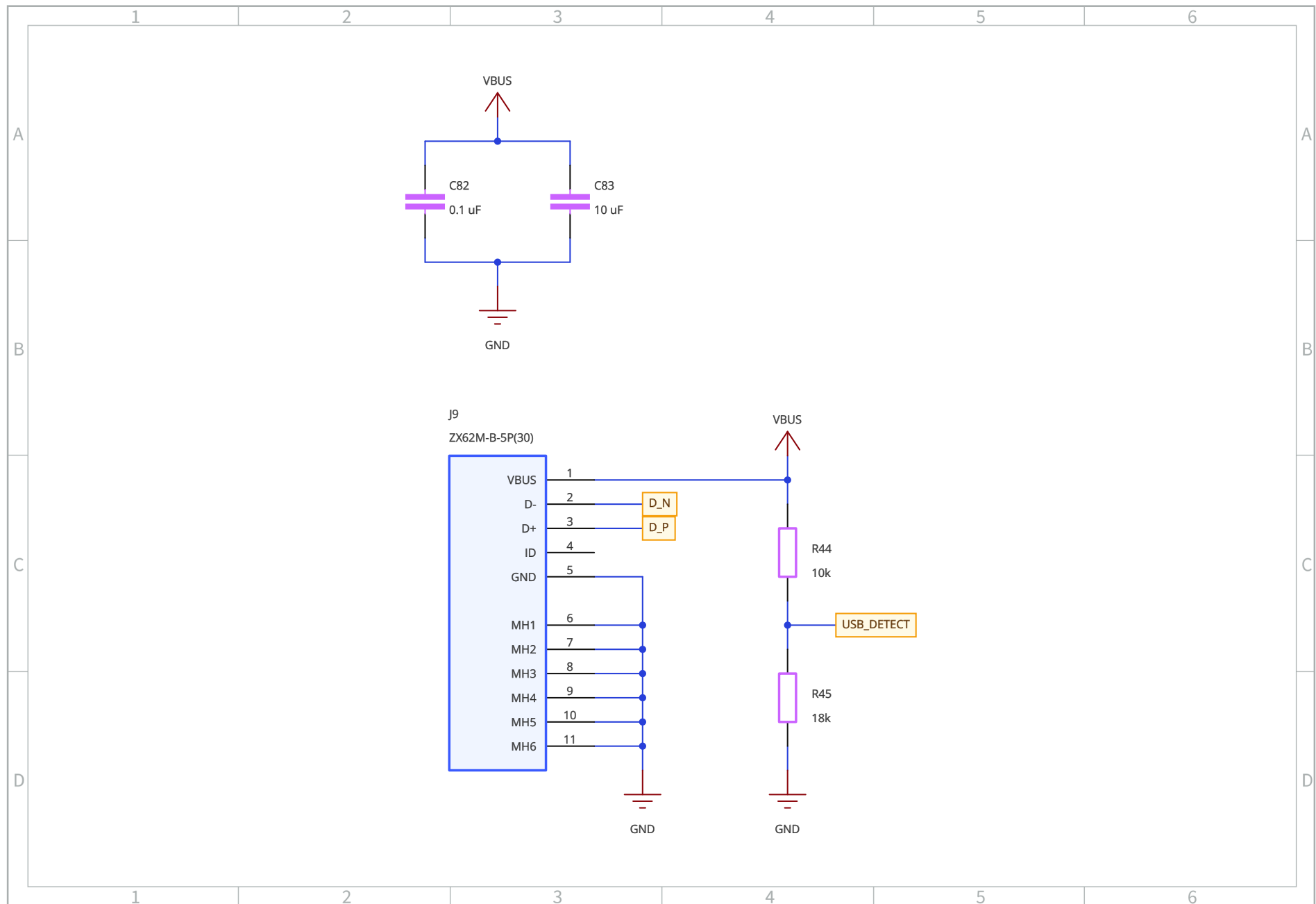


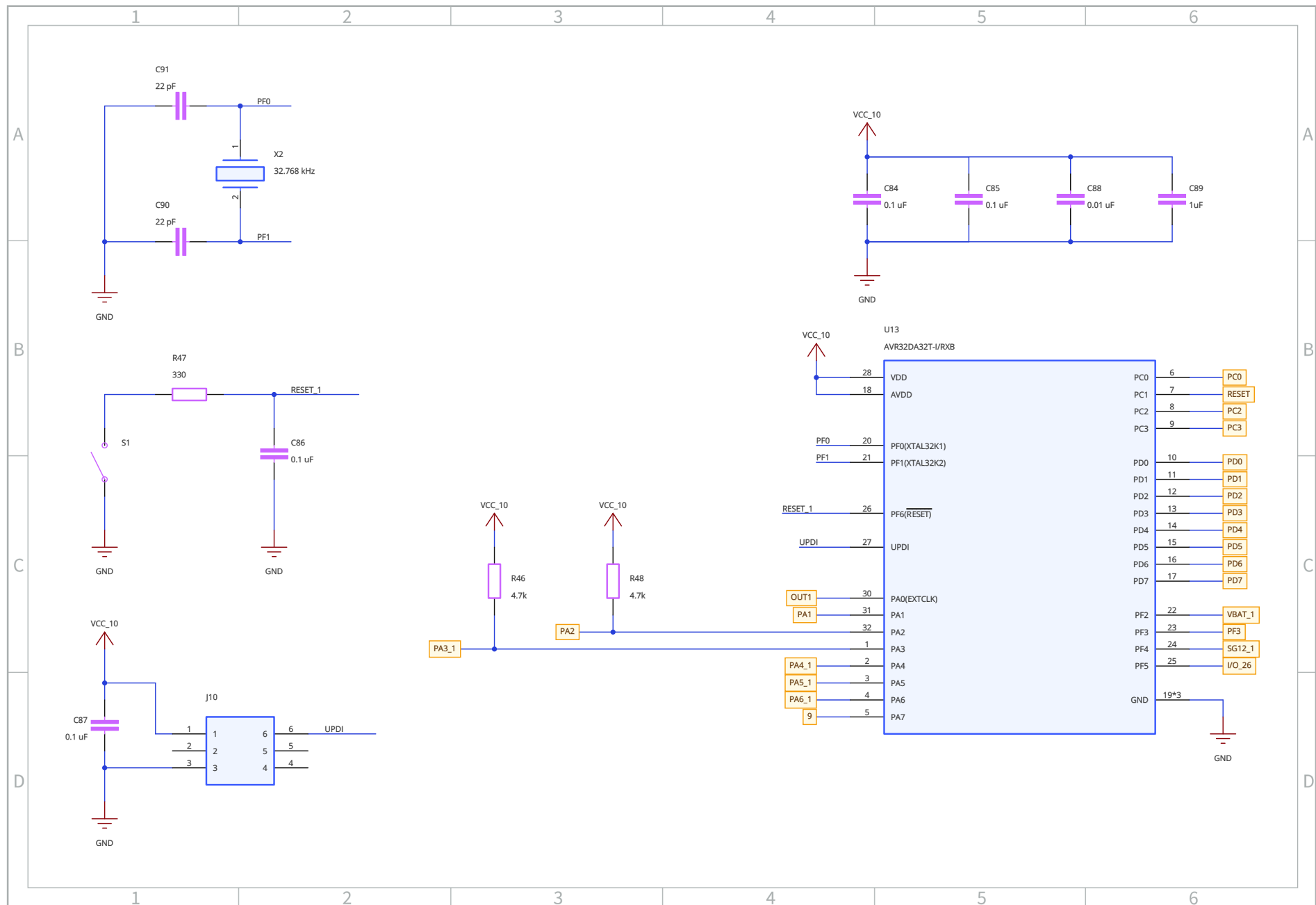


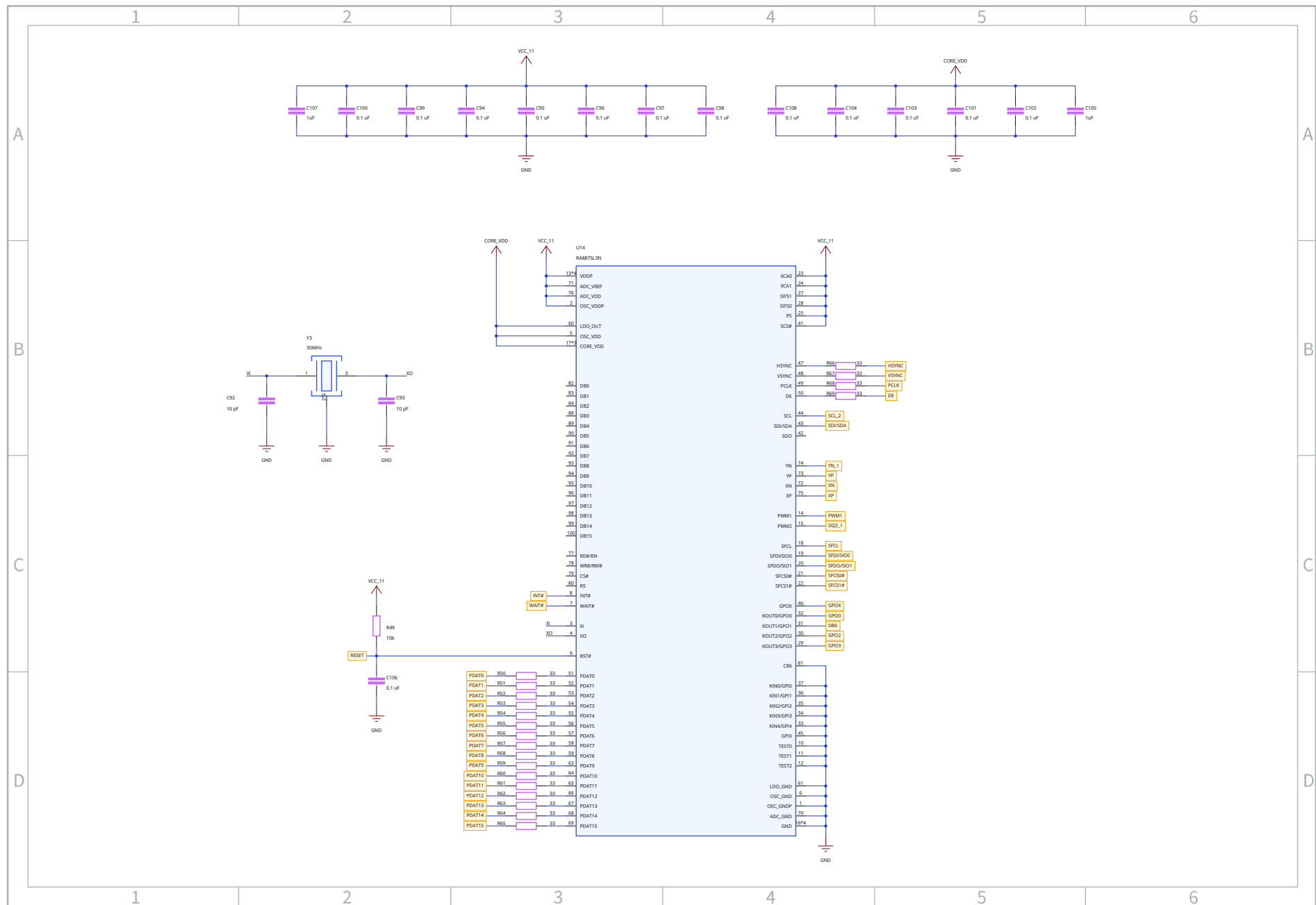




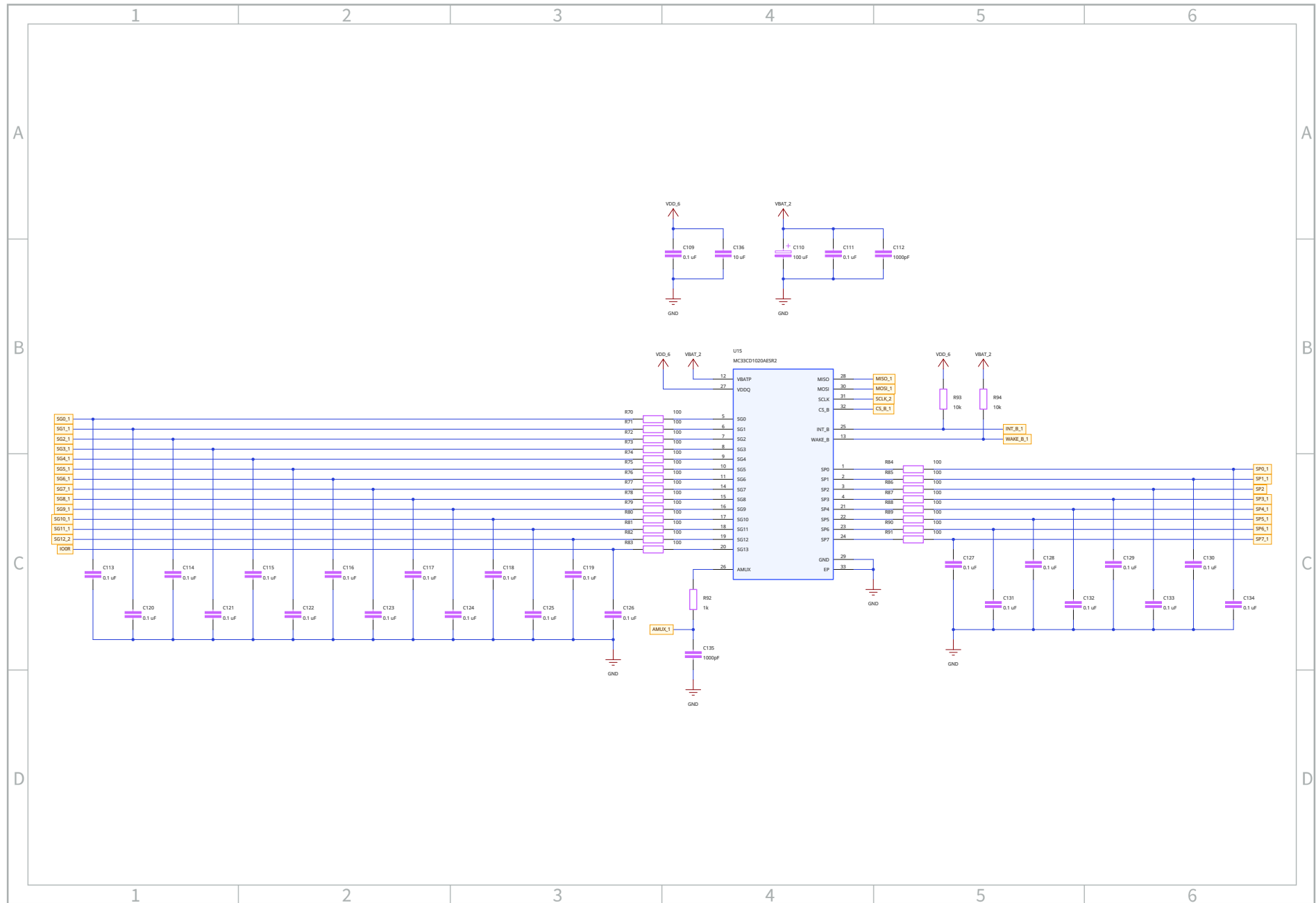


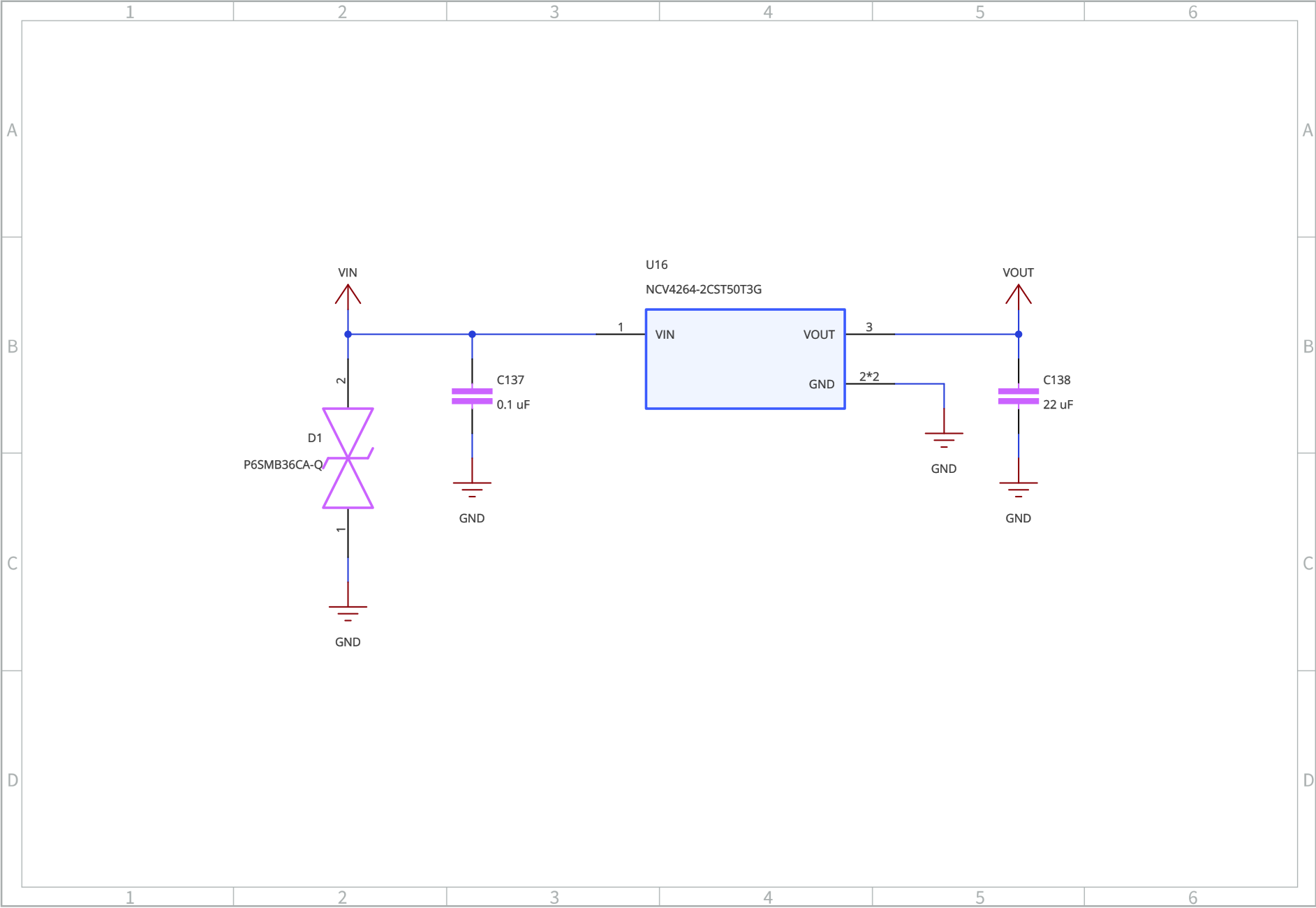


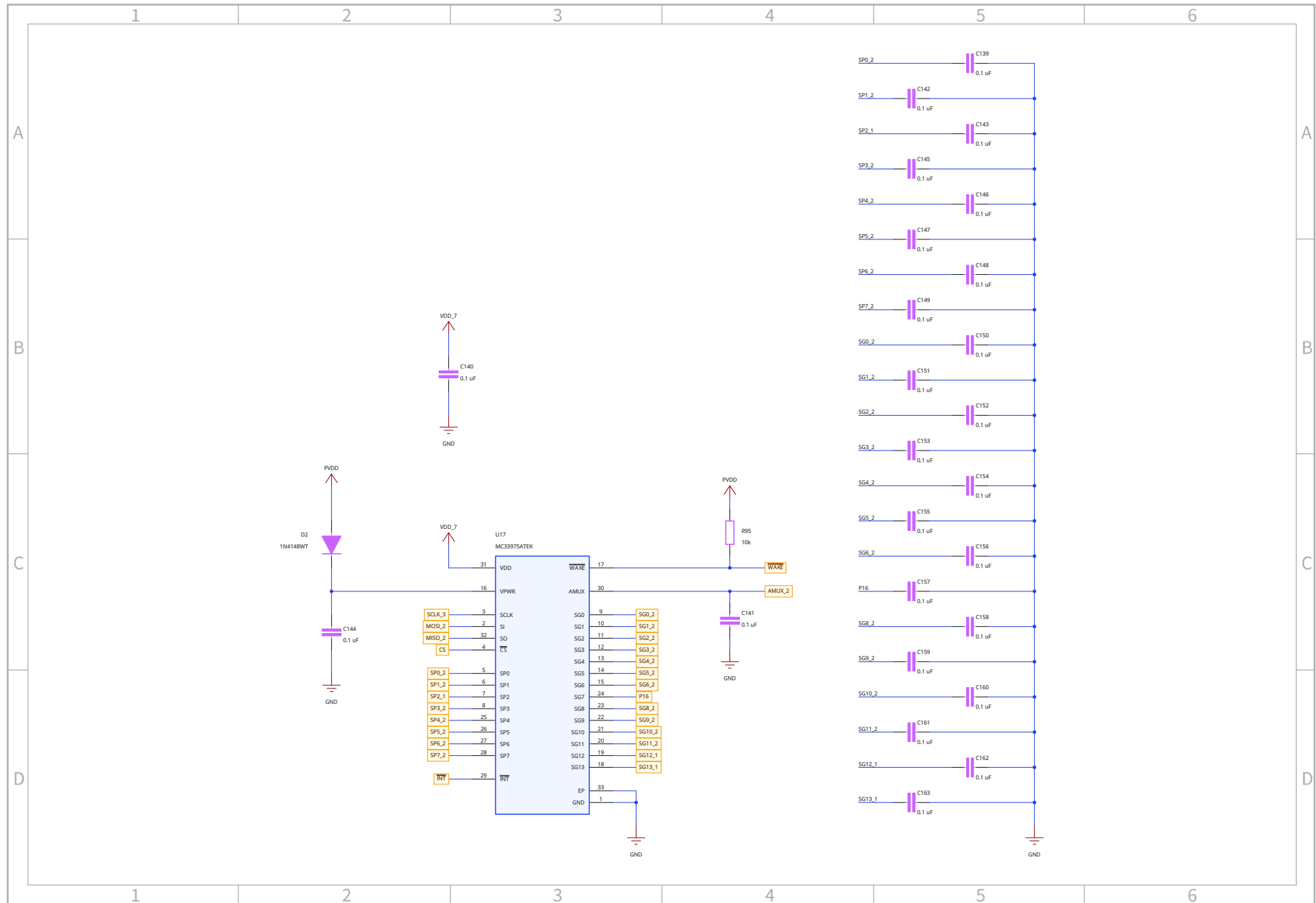




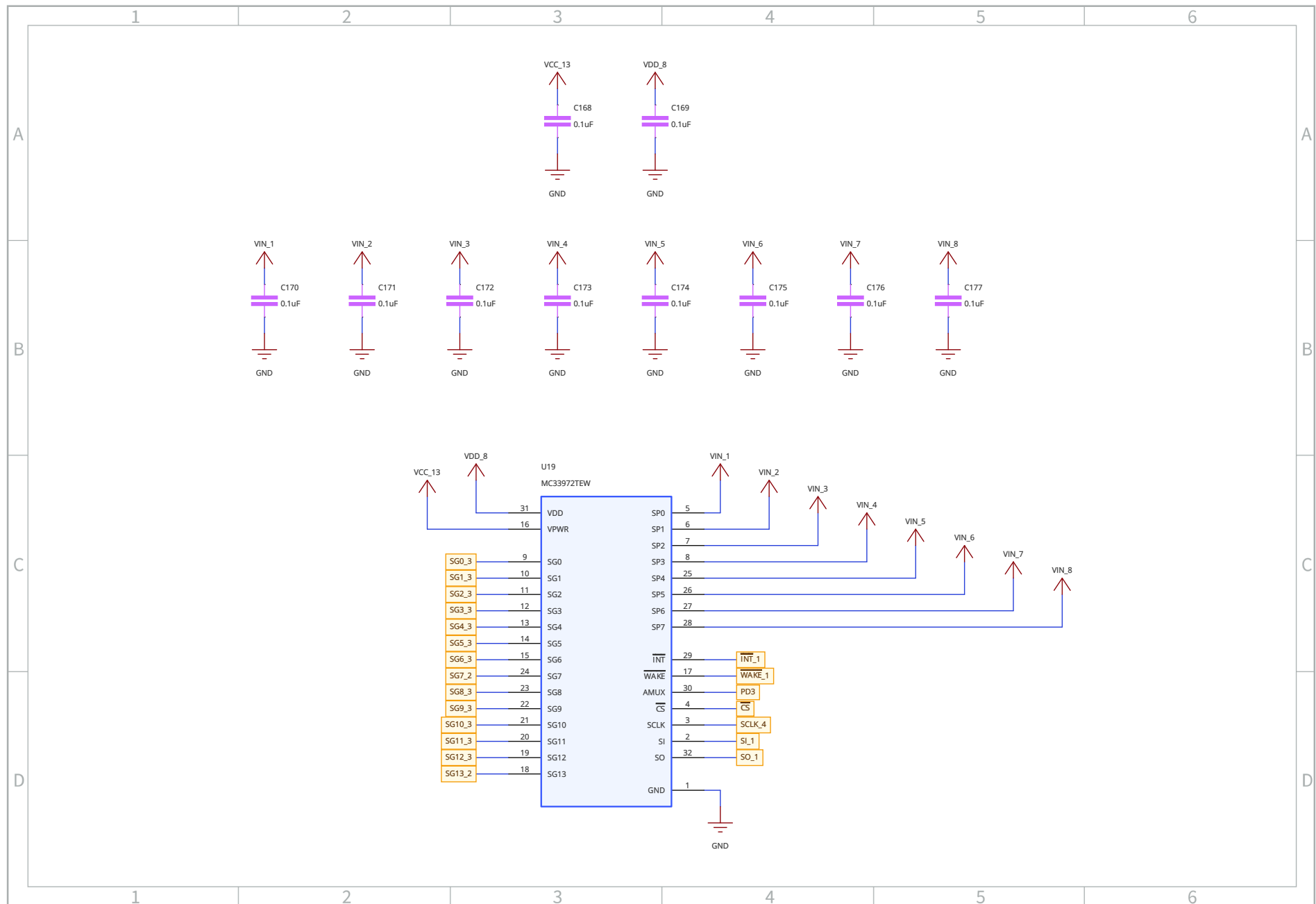


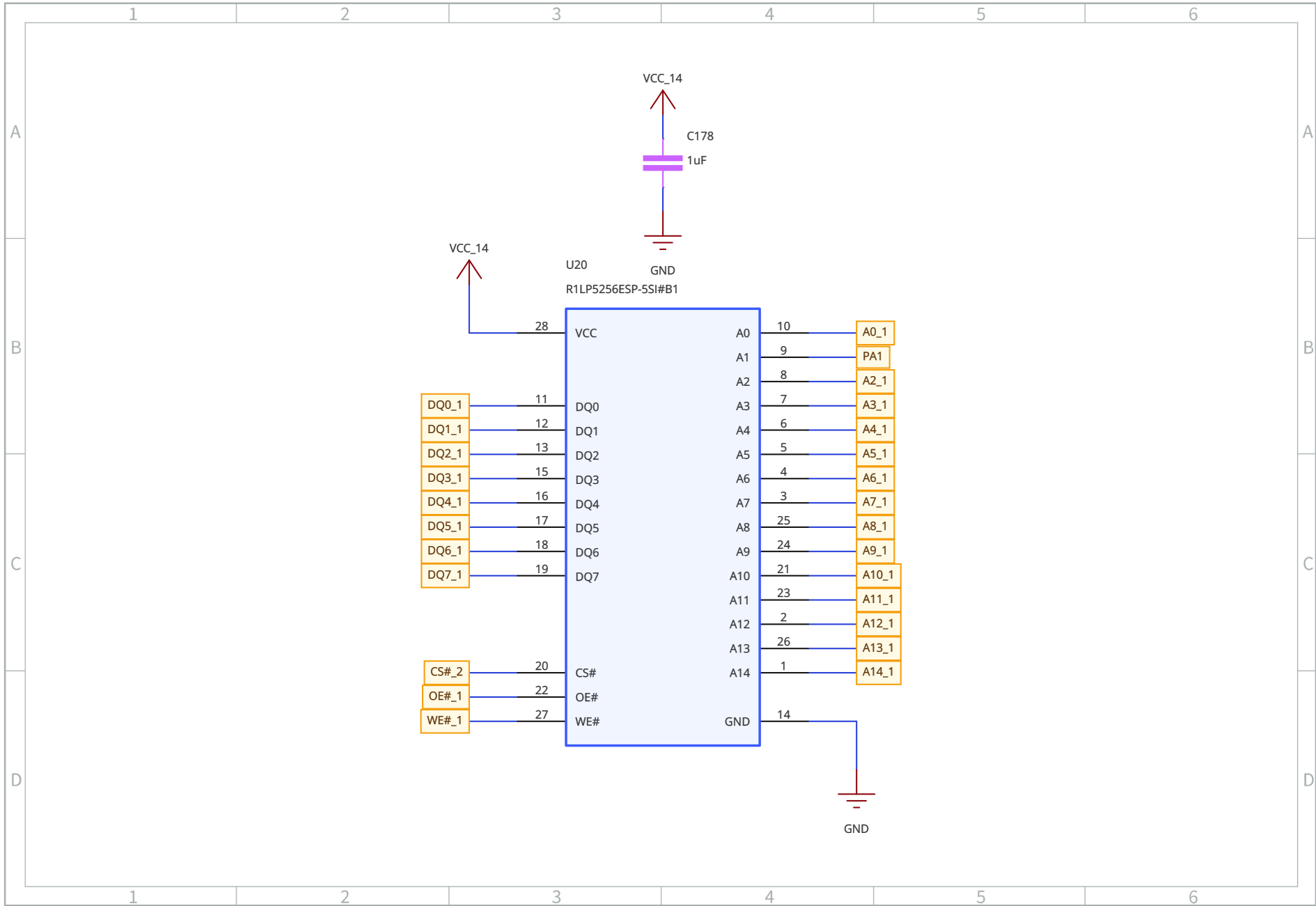


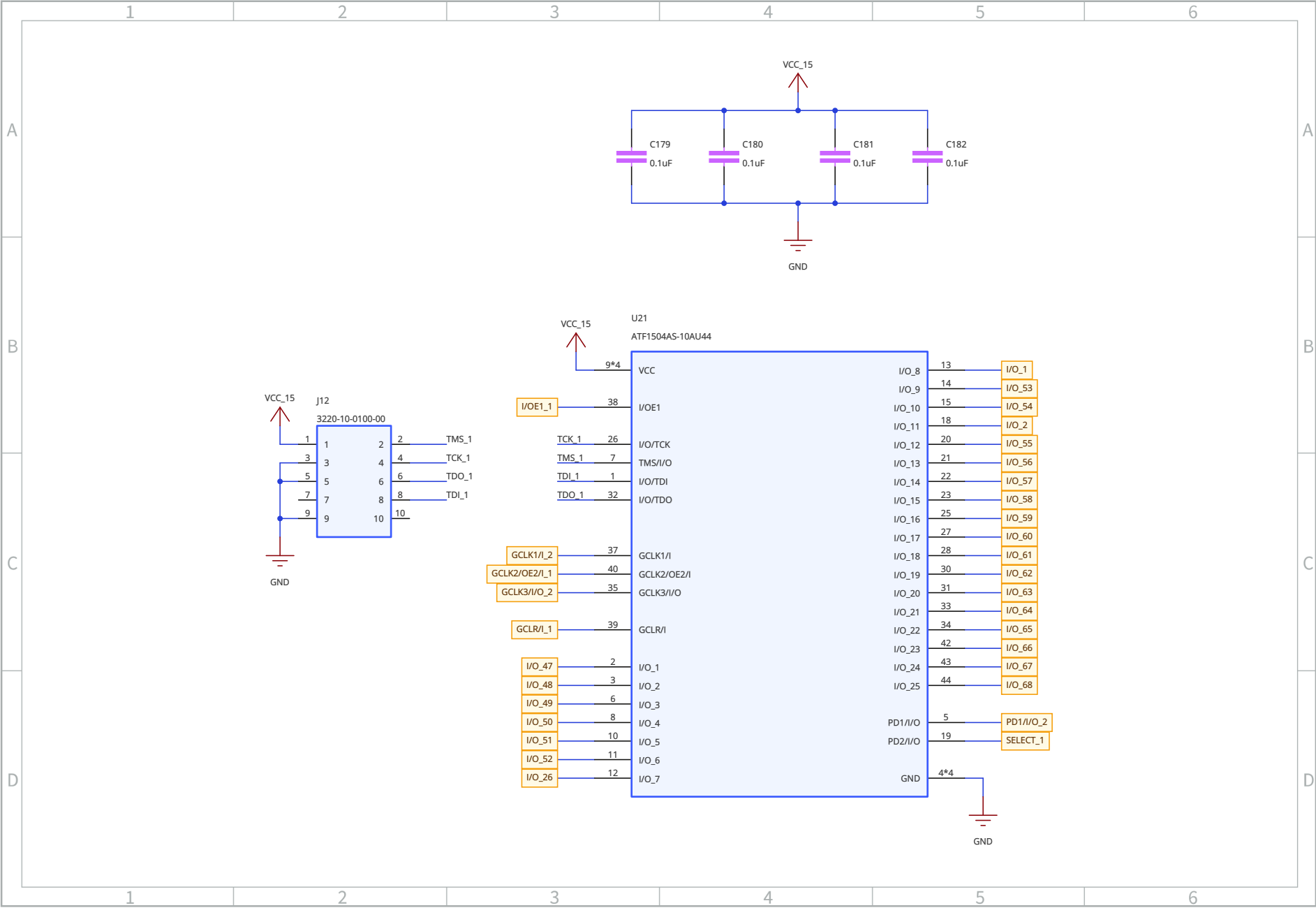


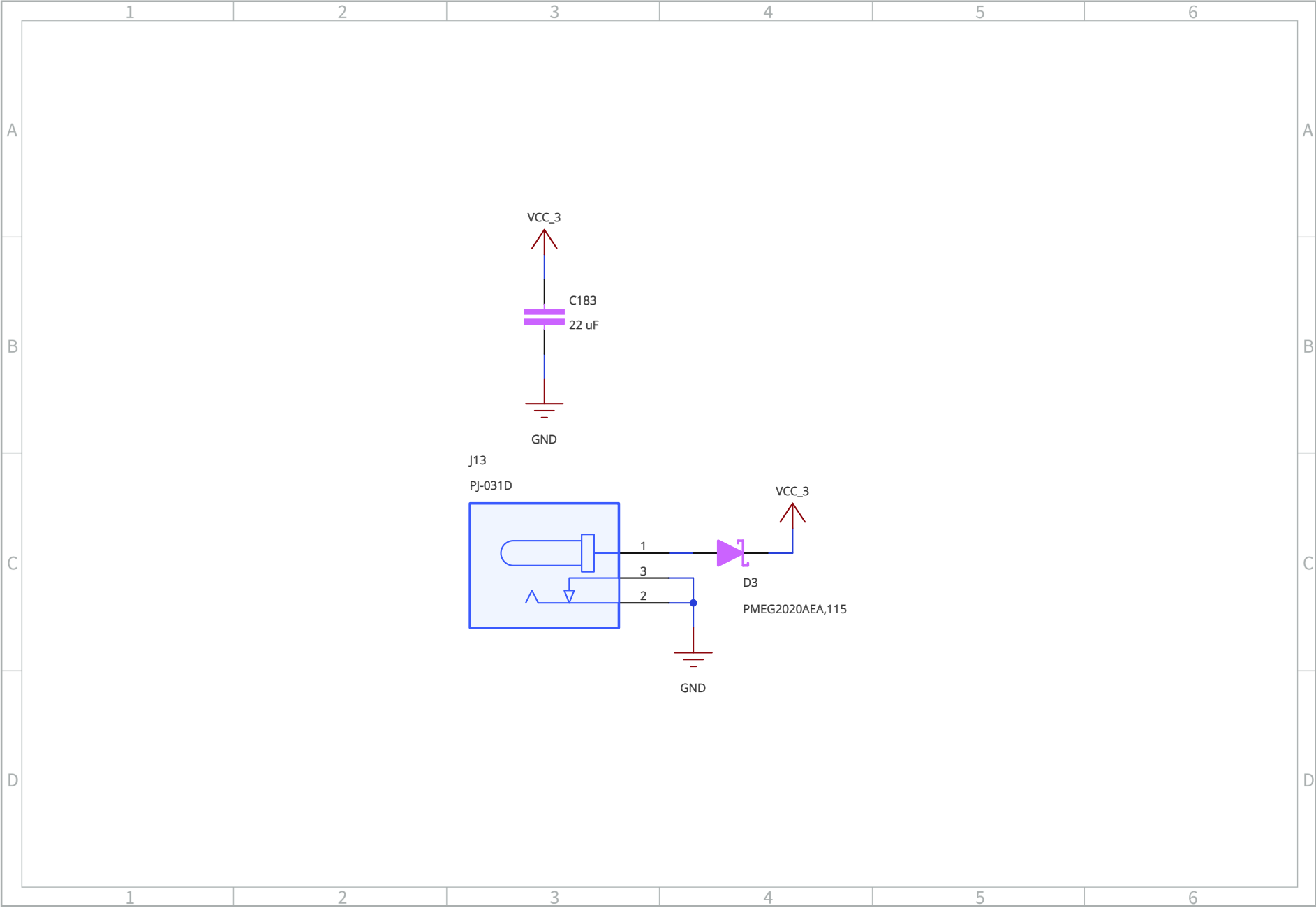




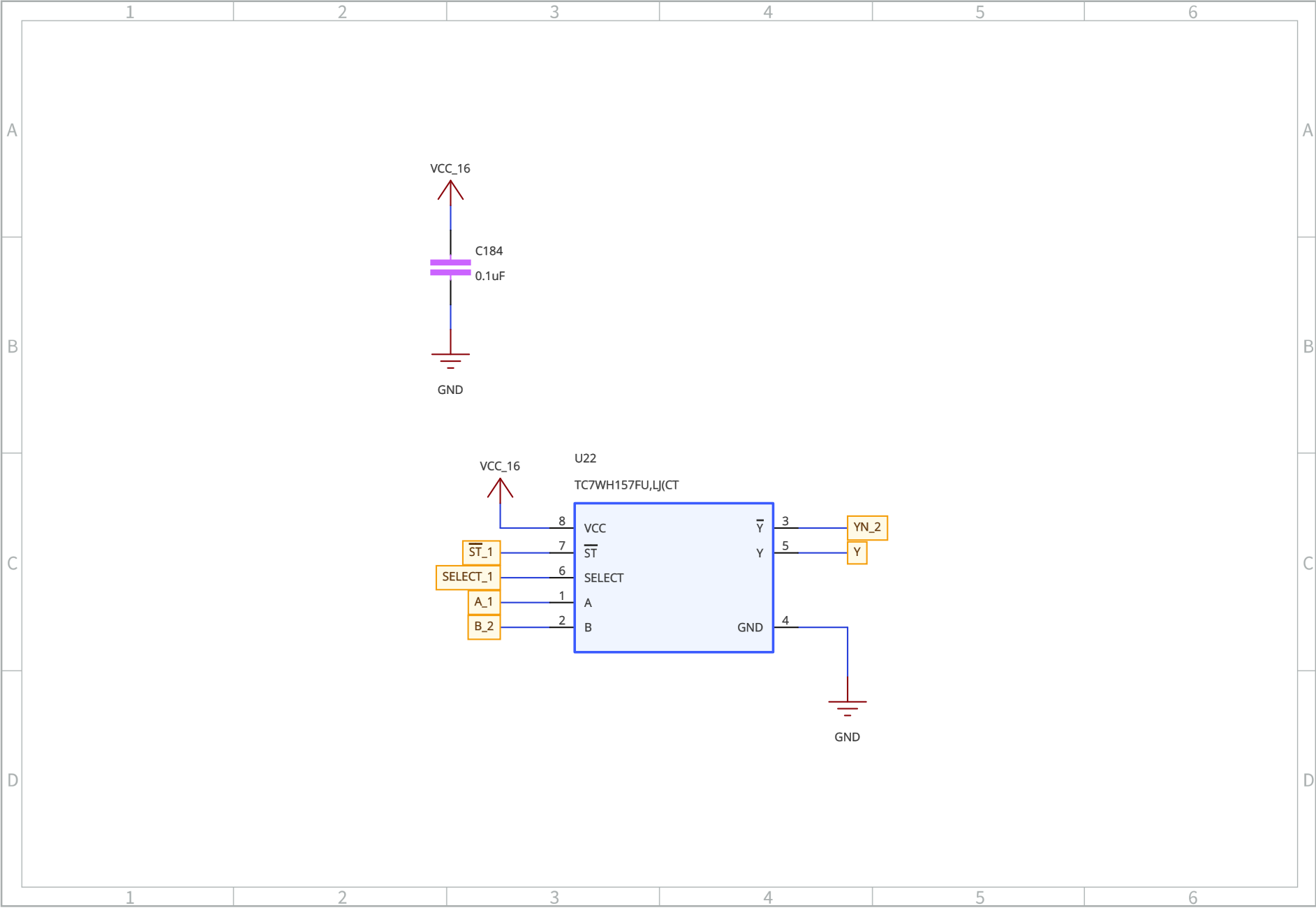












| Ref. Des. | Value  | Manufacturer                   | MPN                | Package    | Description                                                                                              |
|-----------|--------|--------------------------------|--------------------|------------|----------------------------------------------------------------------------------------------------------|
| C1        | 100 nF | MURATA<br>MANUFACTURING CO LTD | GRM033C71C104KE14J | 0603M/0201 | Ceramic Capacitor, YES, Ceramic, 16V, 10% +Tol, 10% -Tol, X7S, 22% TC, 0.1uF, Surface Mount, 0603M/0201  |
| C2        | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C3        | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C4        | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C5        | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C6        | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C7        | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C8        | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C9        | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C10       | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C11       | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C12       | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C13       | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C14       | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C15       | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |

| Ref. Des. | Value  | Manufacturer              | MPN             | Package | Description                                                                                              |
|-----------|--------|---------------------------|-----------------|---------|----------------------------------------------------------------------------------------------------------|
| C16       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KP5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C17       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KP5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C18       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KP5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C19       | 10 µF  | SAMSUNG ELECTRO-MECHANICS | CL05A106MP6NUN8 | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 20% +Tol, 20% -Tol, X5R, 15% TC, 10uF, Surface Mount, 0402  |
| C20       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C21       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C22       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C23       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C24       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C25       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C26       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C27       | 1 µF   | SAMSUNG ELECTRO-MECHANICS | CL05A105KP5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402   |
| C28       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C29       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05A104KO5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C30       | 1 µF   | SAMSUNG ELECTRO-MECHANICS | CL05A105KO5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402   |

| Ref. Des. | Value  | Manufacturer                | MPN                 | Package    | Description                                                                                                       |
|-----------|--------|-----------------------------|---------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| C31       | 18 pF  | KEMET ELECTRONICS CORP      | C0402C180J5GACTU    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000018uF, Surface Mount, 0402 |
| C32       | 18 pF  | KEMET ELECTRONICS CORP      | C0402C180J5GACTU    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000018uF, Surface Mount, 0402 |
| C33       | 100 nF | KYOCERA AVX COMPONENTS CORP | 06035C104K4T2A      | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603          |
| C34       | 100 nF | KYOCERA AVX COMPONENTS CORP | 06035C104K4T2A      | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603          |
| C35       | 100 nF | KYOCERA AVX COMPONENTS CORP | 06035C104K4T2A      | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603          |
| C36       | 4.7 µF | TDK CORP                    | C2012X5R1H475K125AB | 0805       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 4.7uF, Surface Mount, 0805          |
| C37       | 100 µF | PANASONIC CORP              | EEEFTH101XAP        | 2626       | Aluminum Electrolytic Capacitor, Polarized, Aluminum (wet), 50V, 20% +Tol, 20% -Tol, 100uF, Surface Mount, 2626   |
| C38       | 100 µF | PANASONIC CORP              | EEEFTH101XAP        | 2626       | Aluminum Electrolytic Capacitor, Polarized, Aluminum (wet), 50V, 20% +Tol, 20% -Tol, 100uF, Surface Mount, 2626   |
| C39       | 100 nF | YAGEO CORP                  | CC0402KRX5R9BB104   | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C40       | 1 nF   | MURATA MANUFACTURING CO LTD | GRM155R71H102KA01D  | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.001uF, Surface Mount, 1005M/0402         |
| C41       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KA5NNNC     | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C42       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KA5NNNC     | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C43       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KA5NNNC     | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C44       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KA5NNNC     | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C45       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KA5NNNC     | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |

| Ref. Des. | Value  | Manufacturer              | MPN             | Package | Description                                                                                              |
|-----------|--------|---------------------------|-----------------|---------|----------------------------------------------------------------------------------------------------------|
| C46       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C47       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C48       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C49       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C50       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C51       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C52       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C53       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C54       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C55       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C56       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C57       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C58       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C59       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C60       | 100 nF | SAMSUNG ELECTRO-MECHANICS | CL05B104KA5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |

| Ref. Des. | Value  | Manufacturer                | MPN                | Package    | Description                                                                                                       |
|-----------|--------|-----------------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------------|
| C61       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KA5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C62       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KA5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C63       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KA5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C64       | 1 nF   | MURATA MANUFACTURING CO LTD | GRM155R71H102KA01D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.001uF, Surface Mount, 1005M/0402         |
| C65       | 1 µF   | MURATA MANUFACTURING CO LTD | GRM155R61E105MA12D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 20% +Tol, 20% -Tol, X5R, 15% TC, 1uF, Surface Mount, 1005M/0402             |
| C66       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05A104KO5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C67       | 12 pF  | SAMSUNG ELECTRO-MECHANICS   | CL05C120JB5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000012uF, Surface Mount, 0402 |
| C68       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C69       | 12 pF  | SAMSUNG ELECTRO-MECHANICS   | CL05C120JB5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000012uF, Surface Mount, 0402 |
| C70       | 10 µF  | SAMSUNG ELECTRO-MECHANICS   | CL05A106MQ5NUNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 6.3V, 20% +Tol, 20% -Tol, X5R, 15% TC, 10uF, Surface Mount, 0402          |
| C71       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C72       | 10 µF  | SAMSUNG ELECTRO-MECHANICS   | CL05A106MQ5NUNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 6.3V, 20% +Tol, 20% -Tol, X5R, 15% TC, 10uF, Surface Mount, 0402          |
| C73       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C74       | 100 nF | KYOCERA AVX COMPONENTS CORP | KGM05AR51E104KH    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402          |
| C75       | 100 nF | KYOCERA AVX COMPONENTS CORP | KGM05AR51E104KH    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402          |

| Ref. Des. | Value  | Manufacturer                | MPN                | Package    | Description                                                                                                      |
|-----------|--------|-----------------------------|--------------------|------------|------------------------------------------------------------------------------------------------------------------|
| C76       | 100 nF | KYOCERA AVX COMPONENTS CORP | KGM05AR51E104KH    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C77       | 100 nF | KYOCERA AVX COMPONENTS CORP | KGM05AR51E104KH    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C78       | 100 nF | KYOCERA AVX COMPONENTS CORP | 06035C104K4T2A     | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603         |
| C79       | 100 nF | KYOCERA AVX COMPONENTS CORP | KGM05AR51E104KH    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C80       | 1 µF   | SAMSUNG ELECTRO-MECHANICS   | CL10B105KB8NQNC    | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 1uF, Surface Mount, 0603           |
| C81       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C82       | 100 nF | KYOCERA AVX COMPONENTS CORP | KGM05AR51E104KH    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C83       | 10 µF  | SAMSUNG ELECTRO-MECHANICS   | CL10A106KO8NQNC    | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 10uF, Surface Mount, 0603          |
| C84       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C85       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C86       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C87       | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C88       | 10 nF  | SAMSUNG ELECTRO-MECHANICS   | CL05B103KO5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.01uF, Surface Mount, 0402        |
| C89       | 1 µF   | SAMSUNG ELECTRO-MECHANICS   | CL05A105KO5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402           |
| C90       | 22 pF  | MURATA MANUFACTURING CO LTD | GCM1555C1H220JA16D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000022uF, Surface Mount, 1005M/0402 |

| Ref. Des. | Value  | Manufacturer                   | MPN                | Package    | Description                                                                                                      |
|-----------|--------|--------------------------------|--------------------|------------|------------------------------------------------------------------------------------------------------------------|
| C91       | 22 pF  | MURATA<br>MANUFACTURING CO LTD | GCM1555C1H220JA16D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000022uF, Surface Mount, 1005M/0402 |
| C92       | 10 pF  | SAMSUNG ELECTRO-<br>MECHANICS  | CL10C100DB81PNC    | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.00001uF, Surface Mount, 0603 |
| C93       | 10 pF  | SAMSUNG ELECTRO-<br>MECHANICS  | CL10C100DB81PNC    | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.00001uF, Surface Mount, 0603 |
| C94       | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C95       | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C96       | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C97       | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C98       | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C99       | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C100      | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C101      | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C102      | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C103      | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C104      | 100 nF | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A104KP5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402         |
| C105      | 1 µF   | SAMSUNG ELECTRO-<br>MECHANICS  | CL05A105KO5NNNC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402           |



| Ref. Des. | Value  | Manufacturer                   | MPN             | Package | Description                                                                                                               |
|-----------|--------|--------------------------------|-----------------|---------|---------------------------------------------------------------------------------------------------------------------------|
| C106      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05A104KP5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C107      | 1 µF   | SAMSUNG ELECTRO-MECHANICS      | CL05A105KO5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402                    |
| C108      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05A104KP5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C109      | 100 nF | KYOCERA AVX COMPONENTS CORP    | KGM05AR51E104KH | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C110      | 100 µF | WUERTH ELEKTRONIK GMBH & CO KG | 875115452003    | 3333    | Aluminum Electrolytic Capacitor, Polarized, Aluminum (solid Polymer), 20V, 20% +Tol, 20% -Tol, 100uF, Surface Mount, 3333 |
| C111      | 100 nF | KYOCERA AVX COMPONENTS CORP    | KGM05AR51E104KH | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C112      | 1 nF   | WUERTH ELEKTRONIK GMBH & CO KG | 885012206083    | 0603    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.001uF, Surface Mount, 0603                |
| C113      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C114      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C115      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C116      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C117      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C118      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C119      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402                  |
| C120      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402                  |

| Ref. Des. | Value  | Manufacturer                   | MPN             | Package | Description                                                                                                |
|-----------|--------|--------------------------------|-----------------|---------|------------------------------------------------------------------------------------------------------------|
| C121      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C122      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C123      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C124      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C125      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C126      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C127      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C128      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C129      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C130      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C131      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C132      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C133      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C134      | 100 nF | SAMSUNG ELECTRO-MECHANICS      | CL05B104KB54PNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402   |
| C135      | 1 nF   | WUERTH ELEKTRONIK GMBH & CO KG | 885012206083    | 0603    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.001uF, Surface Mount, 0603 |

| Ref. Des. | Value      | Manufacturer              | MPN             | Package | Description                                                                                              |
|-----------|------------|---------------------------|-----------------|---------|----------------------------------------------------------------------------------------------------------|
| C136      | 10 $\mu$ F | SAMSUNG ELECTRO-MECHANICS | CL10A106K08NQNC | 0603    | Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 10uF, Surface Mount, 0603  |
| C137      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C138      | 22 $\mu$ F | SAMSUNG ELECTRO-MECHANICS | CL05A226MQ5N6J8 | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 6.3V, 20% +Tol, 20% -Tol, X5R, 15% TC, 22uF, Surface Mount, 0402 |
| C139      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C140      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05A104KP5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C141      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05A104KP5NNNC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C142      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C143      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C144      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C145      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C146      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C147      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C148      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C149      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C150      | 100 nF     | SAMSUNG ELECTRO-MECHANICS | CL05B104KB5NNWC | 0402    | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |

| Ref. Des. | Value  | Manufacturer                | MPN                | Package    | Description                                                                                              |
|-----------|--------|-----------------------------|--------------------|------------|----------------------------------------------------------------------------------------------------------|
| C151      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C152      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C153      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C154      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C155      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C156      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C157      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C158      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C159      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C160      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C161      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C162      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C163      | 100 nF | SAMSUNG ELECTRO-MECHANICS   | CL05B104KB5NNWC    | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C164      | 100 nF | MURATA MANUFACTURING CO LTD | GCM155R71E104KE02D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 1005M/0402  |
| C165      | 100 nF | MURATA MANUFACTURING CO LTD | GCM155R71E104KE02D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 1005M/0402  |

| Ref. Des. | Value  | Manufacturer                   | MPN                | Package    | Description                                                                                              |
|-----------|--------|--------------------------------|--------------------|------------|----------------------------------------------------------------------------------------------------------|
| C166      | 100 nF | MURATA<br>MANUFACTURING CO LTD | GCM155R71E104KE02D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 1005M/0402  |
| C167      | 100 nF | MURATA<br>MANUFACTURING CO LTD | GCM155R71E104KE02D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 1005M/0402  |
| C168      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C169      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C170      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C171      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C172      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C173      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C174      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C175      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C176      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C177      | 100 nF | YAGEO CORP                     | CC0402KRX5R9BB104  | 0402       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402 |
| C178      | 1 µF   | MURATA<br>MANUFACTURING CO LTD | GRM155R61E105MA12D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 20% +Tol, 20% -Tol, X5R, 15% TC, 1uF, Surface Mount, 1005M/0402    |
| C179      | 100 nF | MURATA<br>MANUFACTURING CO LTD | GCM155R71E104KE02D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 1005M/0402  |
| C180      | 100 nF | MURATA<br>MANUFACTURING CO LTD | GCM155R71E104KE02D | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 1005M/0402  |

| Ref. Des. | Value  | Manufacturer                       | MPN                   | Package    | Description                                                                                                                      |
|-----------|--------|------------------------------------|-----------------------|------------|----------------------------------------------------------------------------------------------------------------------------------|
| C181      | 100 nF | MURATA MANUFACTURING CO LTD        | GCM155R71E104KE02D    | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 1005M/0402                          |
| C182      | 100 nF | MURATA MANUFACTURING CO LTD        | GCM155R71E104KE02D    | 1005M/0402 | Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 1005M/0402                          |
| C183      | 22 µF  | SAMSUNG ELECTRO-MECHANICS          | CL10A226MO7JZNC       | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 16V, 20% +Tol, 20% -Tol, X5R, 15% TC, 22uF, Surface Mount, 0603                          |
| C184      | 100 nF | KYOCERA AVX COMPONENTS CORP        | 06035C104K4T2A        | 0603       | Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603                         |
| D1        |        | BOURNS INC                         | P6SMB36CA-Q           |            | Trans Voltage Suppressor Diode, 600W, 30.8V V(RWM), Bidirectional, 1 Element, Silicon, DO-214AA                                  |
| D2        |        | DIODES INC                         | 1N4148WT-7            |            | Rectifier Diode, 1 Phase, 1 Element, 0.125A, 80V V(RRM), Silicon                                                                 |
| D3        |        | NEXPERIA                           | PMEG2020AEA,115       |            | Rectifier Diode, Schottky, 1 Phase, 1 Element, 2A, 20V V(RRM), Silicon                                                           |
| DS1       |        | NEWHAVEN DISPLAY INTERNATIONAL INC | NHD-14432WG-BTFH-V#T  |            | Graphic Dot Matrix LCD Display                                                                                                   |
| J1        | FEMALE | SULLINS CONNECTOR SOLUTIONS INC    | PPPC062LJBN-RC        |            | Board Connector, 12 Contact(s), 2 Row(s), Female, Right Angle, 0.1 inch Pitch, Solder Terminal, Locking                          |
| J2        | MALE   | WUERTH ELEKTRONIK GMBH & CO KG     | 61300511021           |            | Board Connector, 5 Contact(s), 1 Row(s), Male, Right Angle, 0.1 inch Pitch, Solder Terminal, Locking, Black Insulator, Plug      |
| J3        | FEMALE | SAMTEC INC                         | SSW-105-02-F-S-RA     |            | Board Connector, 5 Contact(s), 1 Row(s), Female, Right Angle, 0.1 inch Pitch, Solder Terminal, Locking, Socket                   |
| J4        | MALE   | J S T MFG CO LTD                   | S2B-PH-SM4-TB(LF)(SN) |            | Board Connector, 2 Contact(s), 1 Row(s), Male, Right Angle, 0.079 inch Pitch, Surface Mount Terminal, Locking, Natural Insulator |
| J5        | FEMALE | SWITCHCRAFT INC                    | 35RAPC4BHN2           |            | 3 Pole(s), Female, Solder Terminal, Locking, Jack                                                                                |
| J6        | MALE   | MOLEX LLC                          | 0015916062            |            | Board Connector, 6 Contact(s), 2 Row(s), Male, Straight, 0.1 inch Pitch, Surface Mount Terminal, Locking                         |
| J7        | MALE   | WUERTH ELEKTRONIK GMBH & CO KG     | 61201021621           |            | Board Connector, 10 Contact(s), 2 Row(s), Male, Straight, 0.1 inch Pitch, Solder Terminal, Locking, Black Insulator, Receptacle  |

| Ref. Des. | Value    | Manufacturer               | MPN                | Package | Description                                                                                                |
|-----------|----------|----------------------------|--------------------|---------|------------------------------------------------------------------------------------------------------------|
| J8        | MALE     | CNC TECH LLC               | 3220-10-0100-00    |         | Board Connector, 10 Contact(s), 2 Row(s), Male, Straight, Solder Terminal                                  |
| J9        |          | HIROSE ELECTRIC CO LTD     | ZX62M-B-5P(30)     |         | USB Connector                                                                                              |
| J10       | MALE     | MOLEX LLC                  | 0015916062         |         | Board Connector, 6 Contact(s), 2 Row(s), Male, Straight, 0.1 inch Pitch, Surface Mount Terminal, Locking   |
| J11       | MALE     | CNC TECH LLC               | 3220-10-0100-00    |         | Board Connector, 10 Contact(s), 2 Row(s), Male, Straight, Solder Terminal                                  |
| J12       | MALE     | CNC TECH LLC               | 3220-10-0100-00    |         | Board Connector, 10 Contact(s), 2 Row(s), Male, Straight, Solder Terminal                                  |
| J13       |          | CUI DEVICES                | PJ-031D            |         | 24V, 2A, 1.3mm Dia                                                                                         |
| R1        | 2.2 Ohm  | PANASONIC CORP             | ERJ2GEJ2R2X        | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 2.2ohm, 50V, 5% +/-Tol, -100,600ppm/Cel, Surface Mount, 0402 |
| R2        | 10 kOhm  | VISHAY INTERTECHNOLOGY INC | MCS0402MC1002FE000 | 0402    | Fixed Resistor, Thin Film, 0.07W, 10000ohm, 50V, 1% +/-Tol, 50ppm/Cel, Surface Mount, 0402                 |
| R3        | 10 kOhm  | VISHAY INTERTECHNOLOGY INC | MCS0402MC1002FE000 | 0402    | Fixed Resistor, Thin Film, 0.07W, 10000ohm, 50V, 1% +/-Tol, 50ppm/Cel, Surface Mount, 0402                 |
| R4        | 10 kOhm  | VISHAY INTERTECHNOLOGY INC | MCS0402MC1002FE000 | 0402    | Fixed Resistor, Thin Film, 0.07W, 10000ohm, 50V, 1% +/-Tol, 50ppm/Cel, Surface Mount, 0402                 |
| R5        | 1 kOhm   | PANASONIC CORP             | ERJ2RKF1001X       | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 1000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402     |
| R6        | 1 kOhm   | PANASONIC CORP             | ERJ2RKF1001X       | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 1000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402     |
| R7        | 1 kOhm   | PANASONIC CORP             | ERJ2RKF1001X       | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 1000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402     |
| R8        | 10 kOhm  | PANASONIC CORP             | ERJ2GEJ103X        | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402    |
| R9        | 100 kOhm | PANASONIC CORP             | ERJ2RKF1003X       | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402   |



| Ref. Des. | Value    | Manufacturer   | MPN             | Package | Description                                                                                                 |
|-----------|----------|----------------|-----------------|---------|-------------------------------------------------------------------------------------------------------------|
| R10       | 1 MOhm   | PANASONIC CORP | ERJ2RKD1004X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 1000000ohm, 50V, 0.5% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R11       | 100 kOhm | PANASONIC CORP | ERJ2RKF1003X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402    |
| R12       | 10 kOhm  | YAGEO CORP     | RC0402JR-0710KL | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.0625W, 10000ohm, 50V, 5% +/-Tol, 100ppm/Cel, Surface Mount, 0402  |
| R13       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R14       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R15       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R16       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R17       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R18       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R19       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R20       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R21       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R22       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R23       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R24       | 100 Ohm  | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |



| Ref. Des. | Value   | Manufacturer   | MPN             | Package | Description                                                                                                |
|-----------|---------|----------------|-----------------|---------|------------------------------------------------------------------------------------------------------------|
| R25       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R26       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R27       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R28       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R29       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R30       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R31       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R32       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R33       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R34       | 100 Ohm | PANASONIC CORP | ERJ2RKF1000X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 100ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R35       | 1 kOhm  | PANASONIC CORP | ERJ2RKF1001X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 1000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402     |
| R36       | 10 kOhm | YAGEO CORP     | RC0402JR-0710KL | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.0625W, 10000ohm, 50V, 5% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R37       | 10 kOhm | PANASONIC CORP | ERJ2GEJ103X     | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402    |
| R38       | 10 kOhm | PANASONIC CORP | ERJ2GEJ103X     | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402    |
| R39       | 10 kOhm | PANASONIC CORP | ERJ2GEJ103X     | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402    |

| Ref. Des. | Value    | Manufacturer   | MPN             | Package | Description                                                                                                |
|-----------|----------|----------------|-----------------|---------|------------------------------------------------------------------------------------------------------------|
| R40       | 10 kOhm  | PANASONIC CORP | ERJ2GEJ103X     | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402    |
| R41       | 10 kOhm  | PANASONIC CORP | ERJ2GEJ103X     | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402    |
| R42       | 15 kOhm  | PANASONIC CORP | ERJ2RKF1502X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 15000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402    |
| R43       | 35 kOhm  | YAGEO CORP     | RC0402FR-0735KL | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.0625W, 35000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R44       | 10 kOhm  | PANASONIC CORP | ERJ2GEJ103X     | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402    |
| R45       | 18 kOhm  | PANASONIC CORP | ERJ2RKF1802X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 18000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402    |
| R46       | 4.7 kOhm | PANASONIC CORP | ERA2AED472X     | 0402    | Fixed Resistor, Thin Film, 0.063W, 4700ohm, 50V, 0.5% +/-Tol, 25ppm/Cel, Surface Mount, 0402               |
| R47       | 330 Ohm  | PANASONIC CORP | ERJ2RKF3300X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 330ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402      |
| R48       | 4.7 kOhm | PANASONIC CORP | ERA2AED472X     | 0402    | Fixed Resistor, Thin Film, 0.063W, 4700ohm, 50V, 0.5% +/-Tol, 25ppm/Cel, Surface Mount, 0402               |
| R49       | 10 kOhm  | PANASONIC CORP | ERJ2GEJ103X     | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402    |
| R50       | 33 Ohm   | PANASONIC CORP | ERJ2RKF33R0X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R51       | 33 Ohm   | PANASONIC CORP | ERJ2RKF33R0X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R52       | 33 Ohm   | PANASONIC CORP | ERJ2RKF33R0X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R53       | 33 Ohm   | PANASONIC CORP | ERJ2RKF33R0X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |
| R54       | 33 Ohm   | PANASONIC CORP | ERJ2RKF33R0X    | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402       |

| Ref. Des. | Value  | Manufacturer   | MPN          | Package | Description                                                                                          |
|-----------|--------|----------------|--------------|---------|------------------------------------------------------------------------------------------------------|
| R55       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R56       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R57       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R58       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R59       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R60       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R61       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R62       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R63       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R64       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R65       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R66       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R67       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R68       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |
| R69       | 33 Ohm | PANASONIC CORP | ERJ2RKF33R0X | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 33ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402 |

| Ref. Des. | Value   | Manufacturer   | MPN          | Package | Description                                                                                             |
|-----------|---------|----------------|--------------|---------|---------------------------------------------------------------------------------------------------------|
| R70       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R71       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R72       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R73       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R74       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R75       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R76       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R77       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R78       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R79       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R80       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R81       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R82       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R83       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |
| R84       | 100 Ohm | PANASONIC CORP | ERJPA3F1000V | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603 |

| Ref. Des. | Value   | Manufacturer                   | MPN                 | Package | Description                                                                                              |
|-----------|---------|--------------------------------|---------------------|---------|----------------------------------------------------------------------------------------------------------|
| R85       | 100 Ohm | PANASONIC CORP                 | ERJPA3F1000V        | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603  |
| R86       | 100 Ohm | PANASONIC CORP                 | ERJPA3F1000V        | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603  |
| R87       | 100 Ohm | PANASONIC CORP                 | ERJPA3F1000V        | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603  |
| R88       | 100 Ohm | PANASONIC CORP                 | ERJPA3F1000V        | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603  |
| R89       | 100 Ohm | PANASONIC CORP                 | ERJPA3F1000V        | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603  |
| R90       | 100 Ohm | PANASONIC CORP                 | ERJPA3F1000V        | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603  |
| R91       | 100 Ohm | PANASONIC CORP                 | ERJPA3F1000V        | 0603    | Fixed Resistor, Metal Glaze/thick Film, 0.25W, 100ohm, 150V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603  |
| R92       | 1 kOhm  | PANASONIC CORP                 | ERJ2RKF1001X        | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 1000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402   |
| R93       | 10 kOhm | PANASONIC CORP                 | ERJ2GEJ103X         | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402  |
| R94       | 10 kOhm | PANASONIC CORP                 | ERJ2GEJ103X         | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402  |
| R95       | 10 kOhm | PANASONIC CORP                 | ERJ2GEJ103X         | 0402    | Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402  |
| S1        |         | WUERTH ELEKTRONIK GMBH & CO KG | 434123043816        |         | Keypad Switch, 1 Switches, SPST, Momentary, 0.05A, 12VDC, 2.06N, Solder Terminal, Surface Mount-straight |
| SW1       |         | C & K COMPONENTS LLC           | KMR221GLFS          |         | Keypad Switch, 1 Switches, SPST, Momentary, 0.05A, 32VDC, 2.5N, Solder Terminal, Surface Mount-straight  |
| SW2       |         | C & K COMPONENTS LLC           | KMR221GLFS          |         | Keypad Switch, 1 Switches, SPST, Momentary, 0.05A, 32VDC, 2.5N, Solder Terminal, Surface Mount-straight  |
| U1        |         | Macronix                       | MX25U25645GZ4I00-TR |         | Macronix - MX25U25645GZ4I00-TR                                                                           |

| Ref. Des. | Value | Manufacturer             | MPN                | Package | Description                                                                                                   |
|-----------|-------|--------------------------|--------------------|---------|---------------------------------------------------------------------------------------------------------------|
| U2        |       | RENESAS ELECTRONICS CORP | 70V631S12PRFGI     |         | Multi-Port SRAM, 256KX18, 12ns, CMOS, PQFP128                                                                 |
| U3        |       | ONSEMI                   | NV25320DWVLT3G     |         | EEPROM, 4KX8, Serial, CMOS, PDSO8                                                                             |
| U4        |       | Renesas Electronics      | 5P49V60B000NLG2    |         | Renesas Electronics - 5P49V60B000NLG2                                                                         |
| U5        |       | RENESAS ELECTRONICS CORP | R5F102AAASP#50     |         | Microcontroller, 16-Bit, FLASH, RL78 CPU, 24MHz, CMOS, PDSO30                                                 |
| U6        |       | NXP SEMICONDUCTORS       | NX20P5090UKAZ      |         | High-Voltage USB PD Power Switch                                                                              |
| U7        |       | NXP SEMICONDUCTORS       | MC33978AES         |         | Switch Detection Interface, 22-switches, 3.3 V / 5.0 V SPI , QFN32                                            |
| U8        |       | RENESAS ELECTRONICS CORP | R1LP5256ESP-5SI#B1 |         | Standard SRAM, 32KX8, 55ns, CMOS, PDSO28                                                                      |
| U9        |       | MICROCHIP TECHNOLOGY INC | ATTINY84A-SSU      |         | RISC Microcontroller, 8-Bit, FLASH, AVR RISC CPU, 20MHz, CMOS, PDSO14                                         |
| U10       |       | MICROCHIP TECHNOLOGY INC | ATF1504ASL-25AU44  |         | EE PLD, 25ns, 64-Cell, CMOS, PQFP44                                                                           |
| U11       |       | TOSHIBA CORP             | TC7WH157FU,LJ(CT   |         | Multiplexer, LV/LV-A/LVX/H Series, 1-Func, 2 Line Input, 1 Line Output, Complementary Output, CMOS, PDSO8     |
| U12       |       | TEXAS INSTRUMENTS INC    | SN74LV163ARGYR     |         | Binary Counter, LV/LV-A/LVX/H Series, Synchronous, Positive Edge Triggered, 4-Bit, Up Direction, CMOS, PQCC16 |
| U13       |       | MICROCHIP TECHNOLOGY INC | AVR32DA32T-I/RXB   |         | RISC Microcontroller, 8-Bit, FLASH, AVR CPU, 24MHz, CMOS, PQCC28                                              |
| U14       |       | RAIO TECHNOLOGY INC      | RA8875L3N          |         | RAIO TECHNOLOGY INC - RA8875L3N                                                                               |
| U15       |       | NXP SEMICONDUCTORS       | MC33CD1020AESR2    |         | Low-Cost 22-CH Multiple Switch Detect Interface                                                               |
| U16       |       | ONSEMI                   | NCV4264-2CST50T3G  |         | Fixed Positive LDO Regulator, 5V, 0.5V Dropout, CMOS, PDSO4                                                   |

| Ref. Des. | Value      | Manufacturer             | MPN                  | Package | Description                                                                                               |
|-----------|------------|--------------------------|----------------------|---------|-----------------------------------------------------------------------------------------------------------|
| U17       |            | NXP SEMICONDUCTORS       | MC33975ATEK          |         | Switch Detection Interface, 22-switches, 32mA Suppressed Wakeup, 3.3 V / 5.0 V SPI , SOIC 32              |
| U18       |            | MICROCHIP TECHNOLOGY INC | ATF1504AS-10AU44     |         | EE PLD, 10ns, 64-Cell, CMOS, PQFP44                                                                       |
| U19       |            | NXP SEMICONDUCTORS       | MC33972TEW           |         | Switch Detection Interface, 22-switches with Suppressed Wakeup, 3.3 V / 5.0 V SPI , SOIC 32               |
| U20       |            | RENESAS ELECTRONICS CORP | R1LP5256ESP-5SI#B1   |         | Standard SRAM, 32KX8, 55ns, CMOS, PDSO28                                                                  |
| U21       |            | MICROCHIP TECHNOLOGY INC | ATF1504AS-10AU44     |         | EE PLD, 10ns, 64-Cell, CMOS, PQFP44                                                                       |
| U22       |            | TOSHIBA CORP             | TC7WH157FU,LJ(CT     |         | Multiplexer, LV/LV-A/LVX/H Series, 1-Func, 2 Line Input, 1 Line Output, Complementary Output, CMOS, PDSO8 |
| X1        | 32.768 kHz | ABRACON LLC              | ABS05-32.768KHZ-T    |         | Parallel - Fundamental Quartz Crystal, 0.032768MHz Nom                                                    |
| X2        | 32.768 kHz | ABRACON LLC              | ABS05-32.768KHZ-T    |         | Parallel - Fundamental Quartz Crystal, 0.032768MHz Nom                                                    |
| Y1        |            | Kyocera AVX              | CX3225CA25000D0HSSCC |         | Kyocera AVX - CX3225CA25000D0HSSCC                                                                        |
| Y2        | 20 MHz     | ABRACON LLC              | ABLS-20.000MHZ-B-2-T |         | Parallel - Fundamental Quartz Crystal, 20MHz Nom                                                          |
| Y3        | 30 MHz     | ECS INC INTERNATIONAL    | ECS-300-10-36-CKL-TR |         | Parallel - Fundamental Quartz Crystal, 30MHz Nom                                                          |